

**Convective Heat Transfer
Within Spent Fuel Canisters:
An Experimental Laboratory Study**

Technical Report

January, 1981

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Office of Nuclear Waste Isolation

BATTELLE Project Management Division

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BIBLIOGRAPHIC DATA

Keyhani, M., F. A. Kulacki, and R. N. Christensen, 1981. *Convective Heat Transfer Within Spent Fuel Canisters: An Experimental Laboratory Study*, ONWI-229, Office of Nuclear Waste Isolation, Battelle Memorial Institute, Columbus, OH, January.

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Printed in the United States of America
Available from
National Technical Information Service
U.S. Department of Commerce
5285 Port Royal Road
Springfield, VA 22161

NTIS price codes
Printed copy: A05
Microfiche copy: A01

ONWI--229

DE82 015756

ONWI-229
Distribution Category UC-70

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This report was prepared by The Ohio State University under Subcontract E512-03900 with Battelle Project Management Division, Office of Nuclear Waste Isolation under Contract No. DE-AC06-76-RL01830-ONWI with the U.S. Department of Energy. This contract was administered by the Battelle Office of Nuclear Waste Isolation.

ABSTRACT

Heat transfer correlations are developed for a vertical annulus wherein the inner cylinder is electrically heated at a rate scaled to that expected in a typical pressurized or boiling water reactor subassembly. The outer cylinder is maintained at a constant temperature and provisions have been made in the design of the apparatus for future experiments with multiple rod arrays having a specific power generation similarly scaled to that in a fuel rod subassembly. Overall heat transfer data are corrected for thermal radiation in the annulus; and Rayleigh numbers, based on the gap width, span the conduction, transition and turbulent regimes of free convection. A total heat transfer coefficient is also obtained for each run via linearization of the radiative transport across the annulus.

Heat transfer coefficients are obtained with air and helium as the fill media. The range of Rayleigh numbers for the present data is $300 < Ra < 2.3 \times 10^6$. Generally, the energy transferred by radiation across the annulus, as a fraction of total heat transfer, varies with Rayleigh number and fill medium. With air as the fill medium, thermal radiation can account for more than 40% of the heat transfer. With helium, radiation can account for up to one-third of the heat transfer rate.

The results of the present work establish a baseline of comparison of heat transfer coefficients to be obtained with a multiple rod array and for the assessment of the effects of thermal radiation on heat transfer in a fuel rod bundle.

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NOMENCLATURE

A	Area, m^2
c_p	Specific heat at constant pressure, $J/kg-^{\circ}C$
C_r	Thermal conductance for radiation, Eq. (3.12)
D	Diameter, m
F_{i-o}	Radiation view factor, inner to outer cylinder
g	Constant of gravitational acceleration, m/s^2
Gr	Grashof number, $\frac{g\beta}{\nu} \ell^3 \Delta T$
h	Convective heat transfer coefficient, $W/m^2-^{\circ}C$
H	Aspect ratio of cylinder, L/ℓ
k	Thermal conductivity, $W/m-^{\circ}C$
K	Radius ratio, r_o/r_i
ℓ	Length scale, $r_o - r_i$, m
L	Height of annulus or layer, m
Nu	Nusselt number based on gap, $h\ell/k$
Nu_L	Nusselt number based on height, hL/k
P	Pressure, atm

Pr	Prandtl number, $\frac{\nu}{\alpha}$
q	Heat flux based on linear cylinder, W/m^2
r	Radius, m
Ra	Rayleigh number, $\text{Gr Pr} = \text{Ra}$
Ra_L	Rayleigh number based on height of annulus, $\frac{g\beta}{\nu\alpha} L^3 \Delta T$
T	Temperature, °K
T_f	Mean temperature of the medium, °K
ΔT	Characteristic temperature difference, °C
W	Power, W

Greek Symbols

α	Thermal diffusivity, $k/\rho c_p$, m^2/s
β	Isobaric coefficient of thermal expansion, $^\circ\text{K}^{-1}$
ϵ	Emissivity
ν	Kinematic viscosity, m^2/s
ρ	Density, kg/m^3
σ	Stephan-Boltzman constant, 5.669×10^{-8} , $\text{W/m}^2\text{-K}^4$

Subscripts

c	Cylinder
i	Inner
m	Mean value
mC	Mean value on cylinder
mR	Mean value on rod
o	Outer
r	Radiation
R	Rod

SECTION 1. INTRODUCTION

Heat transfer within a spent fuel canister is an area where engineering correlations are not presently available to the extent required for the design and analysis of a spent fuel repository. This is due, in part, to the complex internal geometry of the fuel rod assembly and to the variety of thermal boundary conditions that may exist on the outer surface of the canister and in the sleeve between the canister and the host rock of the repository. In the development of waste acceptance criteria for spent fuel and in the design of a spent fuel repository, the maximum temperature within the canister may have to be specified. In addition, the maximum surface temperature of the canister or the temperature gradient within the near-field region of the repository room may also be specified as part of the waste acceptance criteria [5,20]*. The setting of these temperature limits will require detailed knowledge of the convective and the combined convective and radiative heat transfer processes within the canister and fuel rod assembly. The heat transfer processes will, in turn depend on internal canister geometry, specific thermal load (i.e., burnup and age of the fuel at emplacement) and possibly, the type of fill medium within the canister, e.g., gas or otherwise. Related issues concern temperature effects on the corrosion rate of the canister during the period of retrievability and the thermal loading per unit area of the repository. Whatever thermal constraints are set on the design on the repository and the waste package, an important engineering factor will be the heat transfer to the canister from the heat-generating fuel rods.

From a fundamental point of view, the problem of free convection in a vertical cylinder with discrete heat sources, i.e., rods, forms the basis for the computation of heat transfer rates to the surrounding

* Numbers in brackets denote references listed in the Bibliography.

walls and near-field of the repository. Both local and average heat transfer coefficients are of interest, as are temperature distributions as a function of axial location on the rods. In general, heat transfer coefficients in free convection will depend on the overall height-to-diameter ratio of the cylinder, specific and average power generation of the rods, the thermal boundary conditions on the canister surface, and the type of gas or fill medium within the canister. Thus, correlations for the Nusselt number are expected to involve the Grashof number, the Prandtl number, an overall aspect ratio of the cylinder, a rod spacing parameter, and the number of rods and their configuration.

For all but the simplest cases, i.e., the vertical annulus, flow fields are strongly three dimensional, and determining the heat transfer relation requires either an experimental approach or a very costly numerical approach. When effects of thermal radiation between individual rods and between the rods and the canister wall are superimposed upon free convection, the heat transfer problem is made all the more complex via the introduction of nonlinear thermal boundary conditions and surface properties which may be a function of position and time within the canister.

At present, there appears to be only a few studies which treat free convection in fuel rod bundles that may have general applicability to spent fuel disposal. Thermal radiation within fuel rod bundles has been studied both analytically and experimentally in connection with failure of subassembly canisters and cladding in a loss of coolant accident. Taken together, these works point to the need for experimental data and analyses of heat transfer for completely enclosed fuel rod bundles, or subassemblies, over an extended range of Grashof numbers and radiation parameters. Additionally, work is also needed on simpler geometries, e.g., the vertical annulus or a cylinder with a few heated rods to establish fundamental, or limiting-case, heat transfer correlations.

In this report, we describe laboratory experiments on heat transfer in a vertical annulus from the conduction to the turbulent regimes of free convection. Radiation between the walls of the annulus is

generally included in the measurements of the overall heat transfer coefficient at the inner cylinder. Heat transfer coefficients for free convection alone are obtained by applying a correction for the radiative component of transfer obtained by measurements at low pressure. The specific heat generation rate of the inner cylinder is scaled to that expected in a typical pressurized or boiling water reactor fuel rod bundle.



SECTION 2. LITERATURE REVIEW

The heat transfer literature appears to contain little information on free convection in rod bundles which can be applied to spent fuel canister design. Some information exists on heat transfer in a vertical annulus in the multi-cellular flow regime [3], and this will provide only a bounding estimate of Nusselt numbers in the case of a bundle of heated rods. Published work in the nuclear engineering literature has only recently addressed the problem, but no generalization heat transfer correlations have apparently been developed either for a full scale spent fuel canister or a simplified, thermally equivalent model.

The recent study of Wurm [26] includes an extensive literature review on the problem of spent fuel storage and disposal and a presentation of new conceptual designs for the disposal of spent fuel assemblies from pressurized water reactors (PWR's) in a repository in salt. He presents a method for packaging and handling the spent fuel as well. His waste package design includes the emplacement of one spent fuel assembly in a cylinder of carbon steel filled with pressurized helium. For increased mechanical strength, this cylinder can be placed in another cylinder of carbon steel, termed the "overpack". In either form, the package is placed in another cylinder of carbon steel, which serves as the liner of the hole drilled into the salt. The liner protects the package from the salt and moisture which may be released when a temperature gradient is established. To limit the exposure rate in the repository room to less than 2 mrem/h, Wurm suggests the emplacement of a 10 cm thick lead plug on top of the liner, and then burying the entire package 110 cm below the repository room floor. For the distance between each package, it is suggested that, for ten year old spent fuel (739 w/assembly), a separation of 2.8 m be used in a repository room 5.5 m x 5.5 m x 171 m, 600 m below the earth's surface.

This will keep areal power density at 158 kW/acre if the rooms are separated by 13.5 m between their centerlines. For steady-state, one-dimensional heat transfer from a package, with 250°F at the outside surface of the liner, Wurm indicates that the centerline of the assembly experiences a temperature of 340°F. Tables 1 and 2 present the details of the package and liner design.

Llewellyn [18] has made a preliminary comparison of the transient temperature distributions experienced in the storage of ten year old spent fuel assemblies (and high-level waste) in salt repositories with areal power density of 150 kW/acre and 3.5 kW/canister. Two-dimensional unit-cell and far-field heat transfer models were employed in obtaining the solutions to the transient heat conduction equation. Time periods of up to 100,000 years were considered in the calculations. He finds that the decay rate of spent fuel always produces a larger average temperature in the canister than does high-level waste for equal initial areal thermal loading. For example, after forty years in the repository, a peak temperature of 337°F was obtained for spent fuel compared to 301°F in high-level waste. In these calculations, heat transfer processes within the waste packages were not considered, and thus peak temperatures represent volumetrically averaged values.

Recently, Benjamin et al. [1] reported the results of a study of spent fuel heat up following a loss of water during pool storage. Calculations were made taking into account the effects of decay time, fuel element design, storage rack design, packing density, drainage level and other variables on the heat-up characteristics of the spent fuel. They find that clad rupture, or melting, following complete loss of water is strongly dependent on the storage configuration and the decay period. Furthermore, they conclude that the minimum prerequisite storage period to preclude clad failure can vary from less than ten days for some storage configurations to years for others. No consideration was given to the disposal problem wherein heat transfer mechanisms are less effective due to the absence of a liquid cooling medium. However, their study points to the detail one must consider in the heat transfer calculations when predictions of individual rod temperatures are needed.

Table 1 Summary of Package Dimensions and
Materials Proposed by Wurm [26].

Item	Specification
Inside radius	15.38 cm
Outside radius	15.88 cm
Top plate thickness	1.4 cm
Bottom plate thickness	0.5 cm
Minimum lifting lug diameter	3.2 cm
Lifting lug length	8.0 cm
Package inside length	408.0 cm
Package outside length	409.9 cm
Package material	low carbon steel
Lifting lug material	low carbon steel
Filler material	helium
Filler pressure	1 atm

Table 2 Summary of Liner Dimensions and
Materials Proposed by Wurm [26].

Item	Specification
Inside radius	16.38 cm
Outside radius	17.88 cm
Bottom plate thickness	1.5 cm
Liner outside length	432.4 cm
Shield plug thickness	10.1 cm
Cap thickness	1.5 cm
Minimum lifting lug diameter	3.2 cm
Lifting lug length	8.0 cm
Connecting stud diameter	0.5 cm
Connecting stud length	5.0 cm
Gasket thickness	0.6 cm
Coating thickness	0.063 cm
Liner material	low carbon steel
Cap material	low carbon steel
Connecting stud material	low carbon steel
Shield plug material	lead
Coating material	nickel

Callahan et al. [2] report results of heat transfer in the waste container-sleeve/salt interface. They consider this system to be homogeneously heated with a diameter of one foot and a length of ten feet situated in a drill hole 18 ft deep in the center of the floor of the repository room. The repository room is taken to be 16 ft high by 18 ft wide, with a distance of 24 ft between each waste package. With 60 ft thick pillars separating each room, the areal thermal loading was 150 kW/acre. As an extreme case, they considered the periphery to be adiabatic, thus allowing no heat removal across the surface of the room. This gave maximum temperatures in the vicinity of the package. The surface of the earth was considered to be at a constant temperature of either 0°F or 70°F, and the remaining boundaries in the domain of the calculations were taken to be adiabatic. A two-dimensional finite-difference solution of the heat conduction problem was carried out with results that waste package centerline temperatures can reach 900°F within six months, and temperature on the surface of the package can reach 230°F within four years. +

Calculations such as Callahan's when compared to those of Llewellyn, illustrate the degree to which simplifying assumptions can affect predictions of peak temperatures and surface temperature of the waste package. Furthermore, these studies indicate that knowledge of the heat transfer coefficients on the interior surface of the canister would enable predictions to be made with appropriate account taken of the thermal coupling between the canister and the repository.

Free convection of a variable property fluid through an open ended fuel rod bundle has been experimentally and theoretically studied by Davis and Perona [6]. They had treated the case of laminar flow and

+ Callahan et al. assumed thermal conductivity of 2.5 Btu/hr-ft-°F for salt and 0.25 Btu/hr-ft-°F for the waste. A larger effective conductivity of the waste, owing to convection effects within the canister, would lower maximum waste temperatures.

pitch-to-diameter ratios of 1.2 to 2.0. Local Nusselt numbers correlated with a modified Graetz number showed a parametric dependence on both pitch-to-diameter and dimensionless flow rate. The latter was inversely proportional to a modified Grashof number defined in terms of the heat flux at the rod surface. Thus, a standard form of correlation for free convection heat transfer was not possible with the Grashof number concealed within the Graetz number. Physically important results were that outlet values of mass flow rate, hence, fluid velocity within the bundle, are strongly dependent on pitch-to-diameter ratio and the length of the bundle. Local Nusselt numbers for the fuel rods showed a decrease when plotted as a function of the modified Graetz number. These results were obtained via an analytical solution of a simplified form of the non-linear conservation equations within the tube bundle. Experiments on a rod bundle containing forty-two rods, each 0.625 in. in diameter and 48.5 in. long, immersed in air confirmed the results of the analysis for a pitch-to-diameter ratios of 1.68 and 2.03. The experimental data, however, were not corrected for radiative transfer between individual tubes.

The more recent study of Koenig and Buchanan [17] involved measurements of maximum temperatures in an electrically heated rod bundle in free convection with radial conduction and thermal radiation. Their experimental apparatus comprised a bundle of sixty-one rods 5.59 mm in diameter wrapped with wire 1.3 mm in diameter. The wire contained a chromel-alumel thermocouple with the hot junction located at one-fourth, one-half, three-fourths, or top of the heated section. The rods were 1.5 m in length and contained 34.3 cm long sheathed nichrome heaters of 3.18 mm in diameter at the centerline. The axial conduction of the rods helped to create the cosine shape power distribution found in actual fuel rods. The rods were placed in a hexagonal container which was 58.17 mm across the outside flats with a 1.0 mm thick wall. Experiments were conducted with air and argon as the working fluids and with either open or closed ends. This illustrated that radiant heat transfer was reduced by approximately one-third. Because of the higher

temperatures involved for the insulated case, the amount of radial heat transfer was even more. Furthermore, it was shown that the heat load at the same temperature would be about twice that for the uninsulated case as compared to the insulated case. Higher temperatures were measured in argon as compared to air. While the radial component of radiative heat would be about the same, the radial component of conduction through the argon would be less due to its lower thermal conductivity. For natural convection, the gas/rod film coefficient would be less for argon due to effects of its thermophysical properties. At 135 W power input, the maximum temperature was 300°C measured at one-half and three-fourths of the heated region. However, with 144 W power input with the container insulated, the maximum temperature was about 470°C. Koenig and Buchanan also calculated values of the pseudo thermal conductivity of the rod bundle. This was obtained by assuming the hexagonal container to be filled with a uniform, internally heated material. They also calculated the heat transfer coefficients at the next-to-center rod and the edge rod. It should be noted that these heat transfer coefficients are system dependent. In general, Koenig and Buchanan found that radial heat transfer, convection, and radiation combined, count for the majority of the heat transferred from the bundle.

Work which complements that of Davis and Perona [6] and Koenig and Buchanan [17] has been presented by Reilly, Chan, Edwards and Kastenberg [21]. These authors considered transient heat removal by thermal radiation only in a hexagonal fuel rod subassembly at high power density. Axial radiative transfer was neglected in their analysis, which considered all fuel rod surfaces to have grey diffuse radiation properties. A Monte Carlo method of analysis was used to determine view factors between rods and subassembly wall. Energy balances were written to enable calculations of the time dependence of cladding and subassembly wall temperatures for the complete loss of convective cooling. The energy balance for a given row of rods included the incident and leaving radiant fluxes to determine the net thermal radiation to any surface within the row. This radiant flux acted as a

time-dependent boundary condition for the temperature variation of the individual rods and subassembly wall. By approximating the radiant flux as a Fourier series and applying Duhamel's principle, the temperature profile could be expressed in terms of a convergent series. It was found that thermal radiation to the subassembly wall can have a significant effect upon the temperature response of both the rods and the subassembly wall. For the decay heat power levels considered (i.e., an increase from 300 W/cm to 350 W/cm at the initial time, 300 W/cm with loss of flow without scram, and an overpower transient from 300 W/cm to either 500 W/cm or 1000 W/cm, this response remains small as long as the cladding does not melt. Prior to clad melting, the outer three rows develop azimuthal temperature gradients, and the subassembly wall provides a heat sink for the transfer of thermal radiation. Transients in these calculations are on the order of 10 s to 100 s due to the high power levels. With respect to the steady free convection and thermal radiation problem in spent fuel disposal, power densities are several orders of magnitude lower, and the questions of cladding integrity does not appear to be an issue. However, the analysis of Reilly et al. for the radiative transport between fuel rods and the subassembly wall may prove useful in the coupled free convection-radiation problem.

Another study of radiative transfer in fuel rod bundles has been performed by Cox [4]. In this work, two rod bundle geometries, a square array on a square pitch and a hexagonal array on an equilateral triangular pitch, were investigated both experimentally and theoretically. Radiative view factors on these two geometries were derived by using Hottel's [14,15,25] crossed string method. Cox also indicated that the formulation of the radiant energy exchange equation in terms of dimensionless variables is more appropriate in the case of known heat generation rates when temperatures are sought. The three major techniques for treating radiative exchange with diffuse grey enclosures were utilized by Cox in handling the steady-state and transient solutions. He showed that the net radiation method [9,16,22] is most convenient for the steady-state problem, while either Gebhart's [13,14] or Hottel's equations are superior for the transient problem.

Cox presented computer programs for obtaining both steady-state and transient solutions for the temperatures of the rods. In the case of uniform heat generation in each rod and uniform surface properties, the steady-state temperature of the central rod was computed. Measurements were made of the steady-state temperatures in two rod bundles of pitch-to-diameter ratios of 1.24 and 1.37. Each bundle contained 217 rods. The energy generation rates were adjusted to yield center tube temperatures of 800°F and 1000°F. In order to minimize the effects of gaseous conduction and natural convection heat transfer between rods, the experiments were conducted in a vacuum. The rods in each bundle were 5.18 m (17 ft) long with a heated region of 120.6 cm (47.5 in.) at the center. Axial conduction in each rod provided the parabolic heat generation distribution typical of actual spent fuel rods. The comparisons of the experimental data and the theoretical results showed poor agreement when constant radiosity was assumed over the entire circumference of the tube. However, when a piecewise continuous radiosity was assumed over 30° segments of the tube circumference, experimental and theoretical results agreed to within seven percent.

As a base case for developing heat transfer correlations for a fuel rod subassembly in a disposal canister, the study of natural convection in a vertical annulus can be considered a logical starting point. The work of de Vahl Davis and Thomas [7], Thomas and de Vahl Davis [23] and Nagendra, Tirunarayanan and Ramachandran [19] provide results for Nusselt numbers in a form suitable for engineering applications.

De Vahl Davis and Thomas [7] have presented finite difference calculations of natural convection between concentric vertical cylinders. They assumed a Boussinesq fluid with $0.5 \leq Pr \leq 5$. The inner cylinder was at a temperature larger than the outer cylinder and both horizontal boundaries were insulated. Laminar, axisymmetric flows were computed with Rayleigh number based on the gap up to 2×10^5 , $1 \leq K \leq 4$ and $1 \leq H \leq 20$. Results were presented for $Pr = 1$. For $Ra < 10^5$, a single convection cell exists in the annulus, while at larger Rayleigh numbers, multi-cellular flow is observed. Local heat

transfer coefficients at the inner wall are found to increase with radius ratio, K , but to decrease with aspect ratio, H . Secondary flows are found to create regions of diminished heat transfer. Overall heat transfer coefficients for $H = 15$ and $K = 1$ are found to agree with the results of Eckert and Carlson [8] for $2 \times 10^4 \leq Ra \leq 10^5$ in a vertical plane layer. At $H = 5$ and $K = 1$, overall Nusselt numbers were somewhat larger than the measurements reported by Emery [10] at $Ra = 2 \times 10^5$, also in a vertical plane layer. A correlation of the form

$$Nu = f(Pr, K) Ra^{0.3} H^{-1/3} \quad (1)$$

was proposed by de Vahl Davis and Thomas to correlate the overall heat transfer coefficients. The function $f(Pr, K)$ is related to the details of the flow and the type of fluid.

It is interesting to note that the correlations given by Eckert and Carlson [8] and by Emery and Chu [11] for vertical plane layers have nearly the same form as Eq. (1). The correlation given by Eckert and Carlson for $Pr = 0.73$ and laminar flow ($Gr \leq 2 \times 10^5$ in their study) is

$$Nu = 0.119 Gr^{0.3} H^{-0.1}, \quad (2)$$

where the width of the layer, W , replaces $R_o - R_i$ as the length scale in both Gr and Nu . Emery and Chu proposed

$$Nu = 0.28 Ra^{0.25} H^{-0.25} \quad (3)$$

for $10^3 \leq Ra \leq 10^7$.

Thomas and de Vahl Davis [23] extended the work of de Vahl Davis and Thomas [7] to provide correlation equations for heat transfer for several ranges of Rayleigh numbers. Using the same boundary conditions as in the previous study, they solved the governing partial differential equations numerically. They found that the flow field is strongly

dependent upon the Rayleigh number and the cavity geometry, whereas the effect of the Prandtl number is small. They obtained correlations for the overall Nusselt numbers for three different flow regimes, namely the conduction regime, transition regime, and the boundary layer regime. The conduction and boundary layer regimes were defined by the following relations:

$$\begin{aligned} Ra/H < 400 \ (H \geq 5, Pr = 1): & \text{Conduction Regime,} \\ Ra/H > 3000 \ (H \geq 5, Pr = 1): & \text{Boundary Layer Regime,} \end{aligned}$$

and

$$\begin{aligned} Ra/H < 1000 \ (H = 1): & \text{Conduction Regime,} \\ Ra/H > 8000 \ (H = 1): & \text{Boundary Layer Regime.} \end{aligned}$$

In all of these cases, the radius ratio, K , was found to weakly affect the limits of the various regimes of flow.

Heat transfer correlations developed by Thomas and de Vahl Davis [23] were,

Conduction Regime:

$$Nu = 0.595 Ra^{0.101} Pr^{0.024} H^{-0.052} K^{0.505} \quad (4)$$

Transition Regime:

$$Nu = 0.202 Ra^{0.294} Pr^{0.097} H^{-0.246} K^{0.423} \quad (5)$$

Boundary Layer Regime:

$$Nu = 0.286 Ra^{0.258} Pr^{0.006} H^{-0.238} K^{0.442} \quad (6)$$

These correlations were developed for $1 \leq K \leq 10$ and $1 \leq H \leq 33$. They also observed that with moderate aspect ratio, the motion takes the form

of a single cell up to a certain Rayleigh number. As the Rayleigh number is increased, this motion becomes unstable and a multicellular mode rapidly develops. The critical Rayleigh number in the conduction regime is given by

$$Ra_{cr} = 2.24 \times 10^7 Pr^{0.5} H^{-2.27} k^{0.88}. \quad (7)$$

When the Rayleigh number is sufficiently high, so that the base flow is no longer in the conduction regime, the mechanism of instability is different. In that case, the critical Rayleigh number is given by

$$Ra_{cr} = 6.0 \times 10^5 Pr^{0.36} H^{-0.96} k^{-0.18}. \quad (8)$$

In Fig. 1 the results of Thomas and de Vahl Davis [23] are summarized to illustrate the Rayleigh number dependence of the several regimes of heat transfer in a vertical annulus. Also plotted are the results of Eckert and Carlson [8] and Emery and Chu [11] for comparison. Generally, when the heat transfer process is well enough removed from the conduction regime, the Nusselt numbers for the small gap vertical annulus and the vertical layer follow nearly the same Rayleigh number dependence in the so called boundary layer regime of flow. This is usually for large Rayleigh numbers, but within the limitation of laminar flow of the several studies.

Free convection in vertical annuli has also been investigated experimentally and theoretically by Nagendra, Tirunarayanan and Ramachandran [19]. They extended the approximate boundary layer model used by Emery and Chu [11] for the case of parallel plates, or vertical layers, to an annulus to obtain heat transfer correlations in the laminar boundary layer regime of convection. They used constant temperature vertical boundaries with the inner cylinder at a higher temperature than the outer cylinder. The major assumptions used by Nagendra et al. in their analysis were, (1) axisymmetric flow prevails, (2) the flow is steady, (3) the fluid is a Boussinesq fluid, (4) laminar boundary layers exist on each cylinder and these do not mix within the annular space, i.e., the gap is large compared to the thickness of the

boundary layers. In their experiments, Nagendra et al. used a stainless steel cylindrical heater, 0.32 in. in diameter, inside a 0.72 in. diameter glass tube. The fluid contained between these cylinders was water. Generally, their heat transfer measurements were in agreement with their analytical results.

Heat transfer correlations reported by Nagendra et al. [19] are short annulus:

$$Nu_L = \frac{0.48 Ra_L}{6830(L/D_o)^4 (D_i/L) + Ra^{3/4}}, \quad (9)$$

where $L/D < 0.1 Ra^{1/4}$;

Long Annulus:

$$Nu_L = \frac{1.19 Ra_L}{16900(L/D_o)^4 (D_i/L) + Ra^{0.84} (D_i/L)^{0.36}} \quad (10)$$

where $0.1 Ra^{1/4} \leq L/D_i \leq 0.738 Ra^{1/4}$;

Thin Wires in Tubes

$$Nu_L = \frac{0.845 Ra_L}{12000(L/D_o)^4 (D_i/L) + Ra^{0.95} (D_i/L)^{0.8}} \quad (11)$$

where $L/D_i > 0.738 Ra^{1/4}$.

When comparing the results of Nagendra et al. [19] to those previously discussed, a redefinition of terms is needed to put the Nusselt and Rayleigh numbers on a consistent basis. This has been done so that Nusselt and Rayleigh numbers are expressed in terms of the height of the annulus, L . This introduces additional factors of the aspect ratio, H , into Eqs. (1) to (8). Based on the dimensions of the experimental apparatus constructed in the present study (see Section 3)

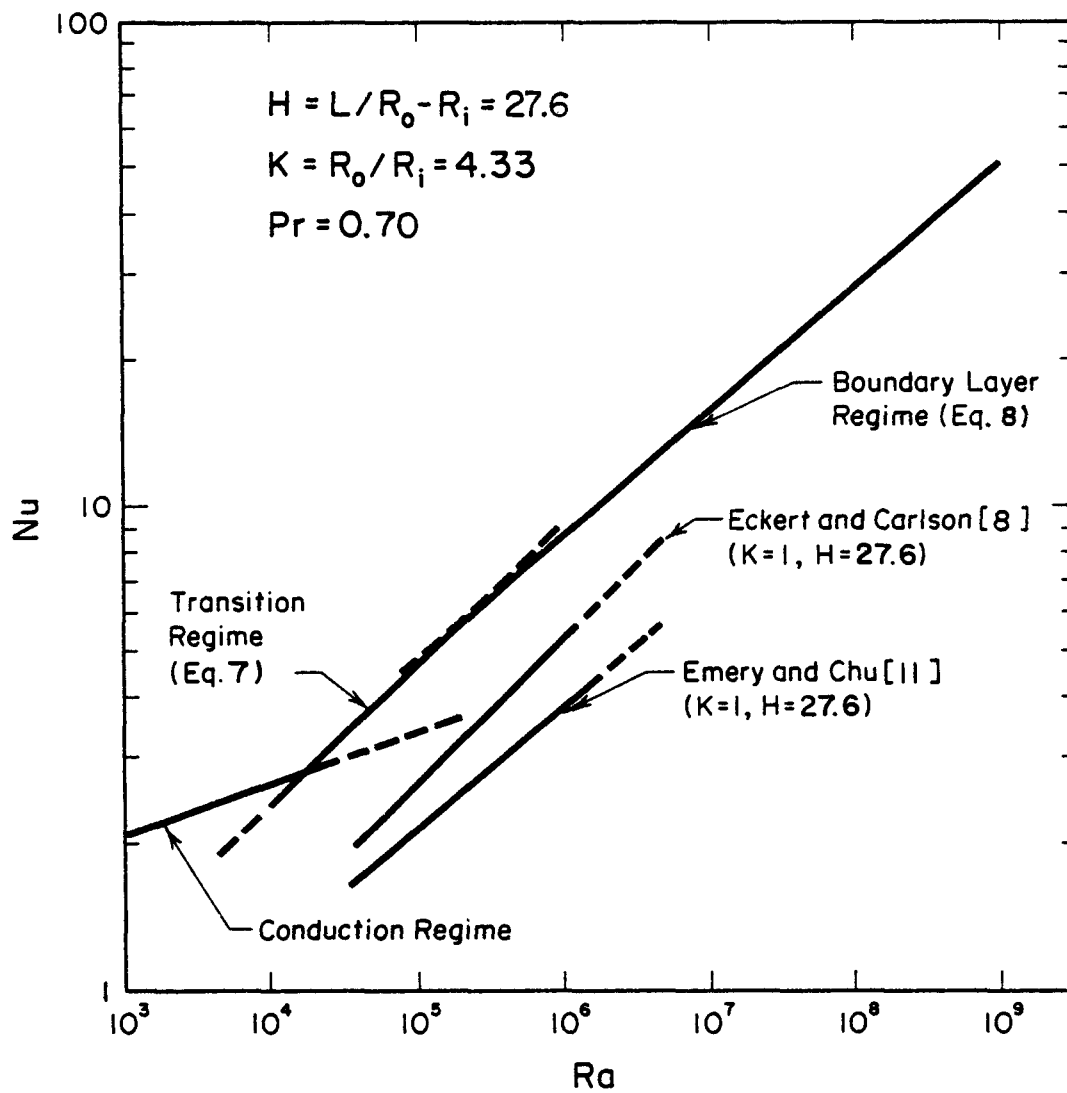


Figure 1. Nusselt Numbers of Free Convection in Vertical Annuli and Rectangular Slots.

and for air ($Pr = 0.7$) as the heat transfer fluid, Fig. 2 has been prepared. For $Ra_L > 10^{10}$, Thomas and de Vahl Davis [23] and Nagendra et al. [19] (Eq. (10)) give results that are virtually identical since they could not be resolved within the accuracy of usual laboratory experiments. At lower Rayleigh numbers, the boundary layer model of Nagendra et al. does not compare well with the numerical calculation of Thomas and de Vahl Davis which is able to follow changes in flow structure at lower Rayleigh numbers.

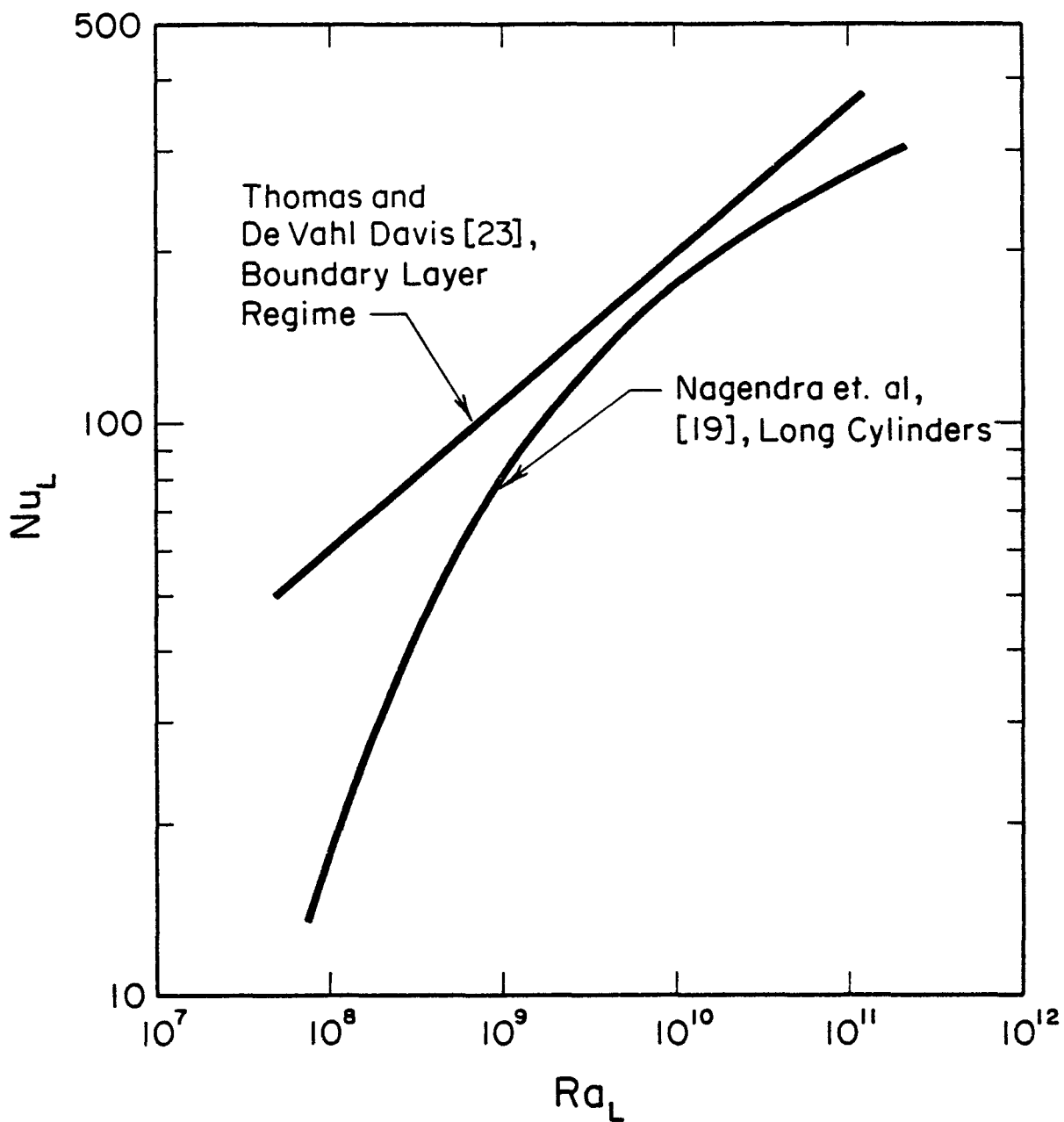


Figure 2. Nusselt Numbers in Free Convection in Long Annuli for the Boundary Layer Regime of Flow. Test Fluid is Air $Pr = 0.7$.

SECTION 3. DESIGN OF THE EXPERIMENT

3.1 Introduction

Heat transfer in spent fuel disposal involves the coupled processes of free convection and thermal radiation inside the canister and combined convection and conduction at the waste package-repository interface. The internal problem is complicated by the close spacing of the heat generating fuel rods, which leads to a non-uniform distributions of heat transfer coefficients and radiosity on each rod within the bundle.

As mentioned in Section 1, the limiting case problem is that of heat transfer in a vertical annulus, for which some data are available. One step removed from this is the determination of heat transfer coefficients in a bundle of several heated rods in a vertical cylinder. In this section, we describe the design and construction of a laboratory apparatus with which to conduct experiments on both of these problems. The apparatus constructed for the present study is nearly a "thermally" equivalent scale model of a spent fuel assembly and canister.

3.2 Basis of the Experimental Design

In order to discuss in some detail the experimental apparatus and the plan of the experiments, it is worthwhile considering some of the details of actual fuel rod assemblies. Figures 3 and 4 present schematics of pressurized (PWR) and boiling water reactor (BWR) fuel rod assemblies, respectively. Table 3 presents typical design specifications of PWR and BWR fuel rod assemblies. In the present study, the experimental apparatus was designed to model a PWR assembly since a greater number of operating reactors are of the PWR type. Furthermore, energy generation by a PWR spent fuel assembly per unit mass of heavy metal is greater than that of a BWR assembly (Table 4),

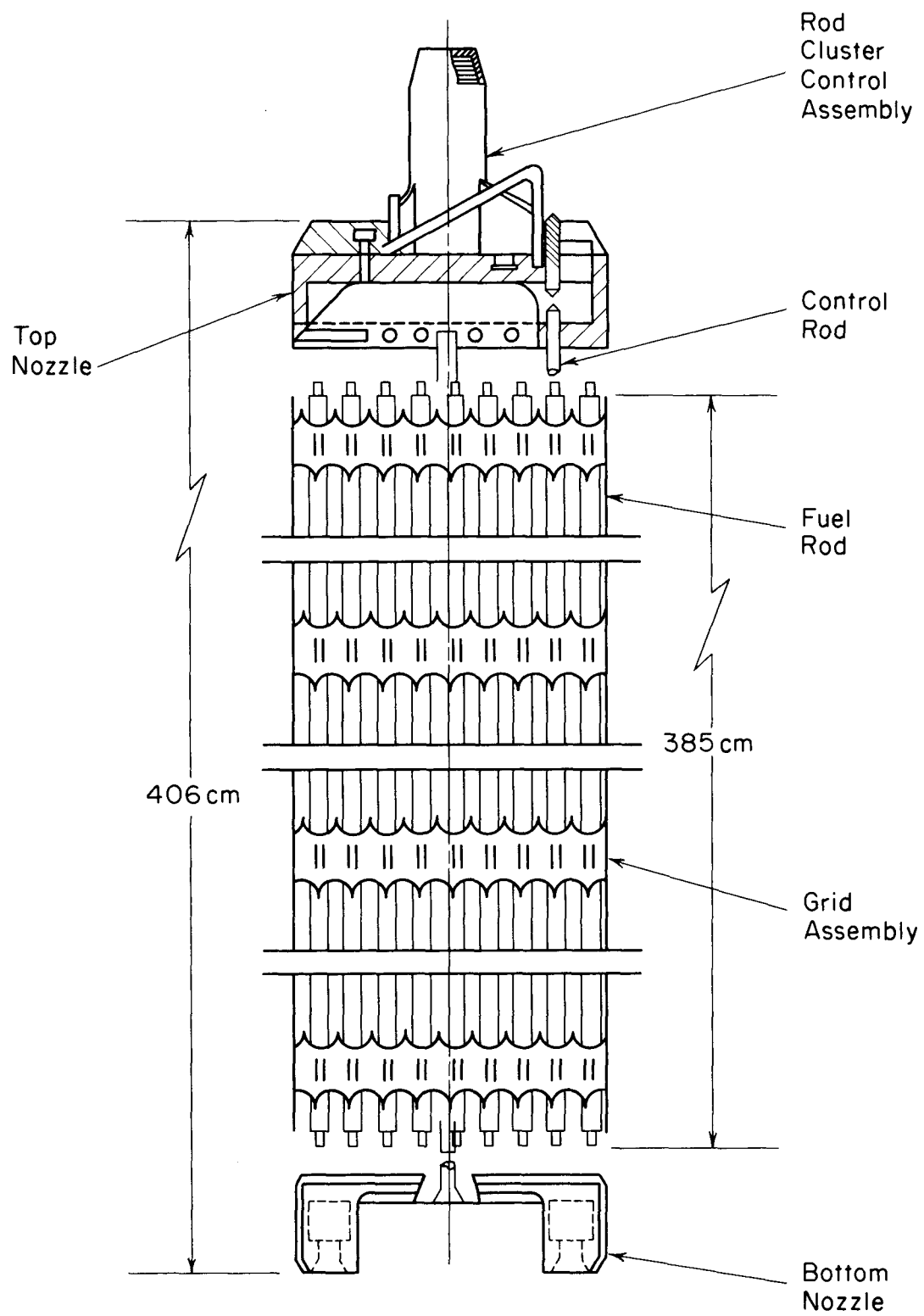


Fig. 3 Pressurized Water Reactor Fuel Rod Assembly [27].

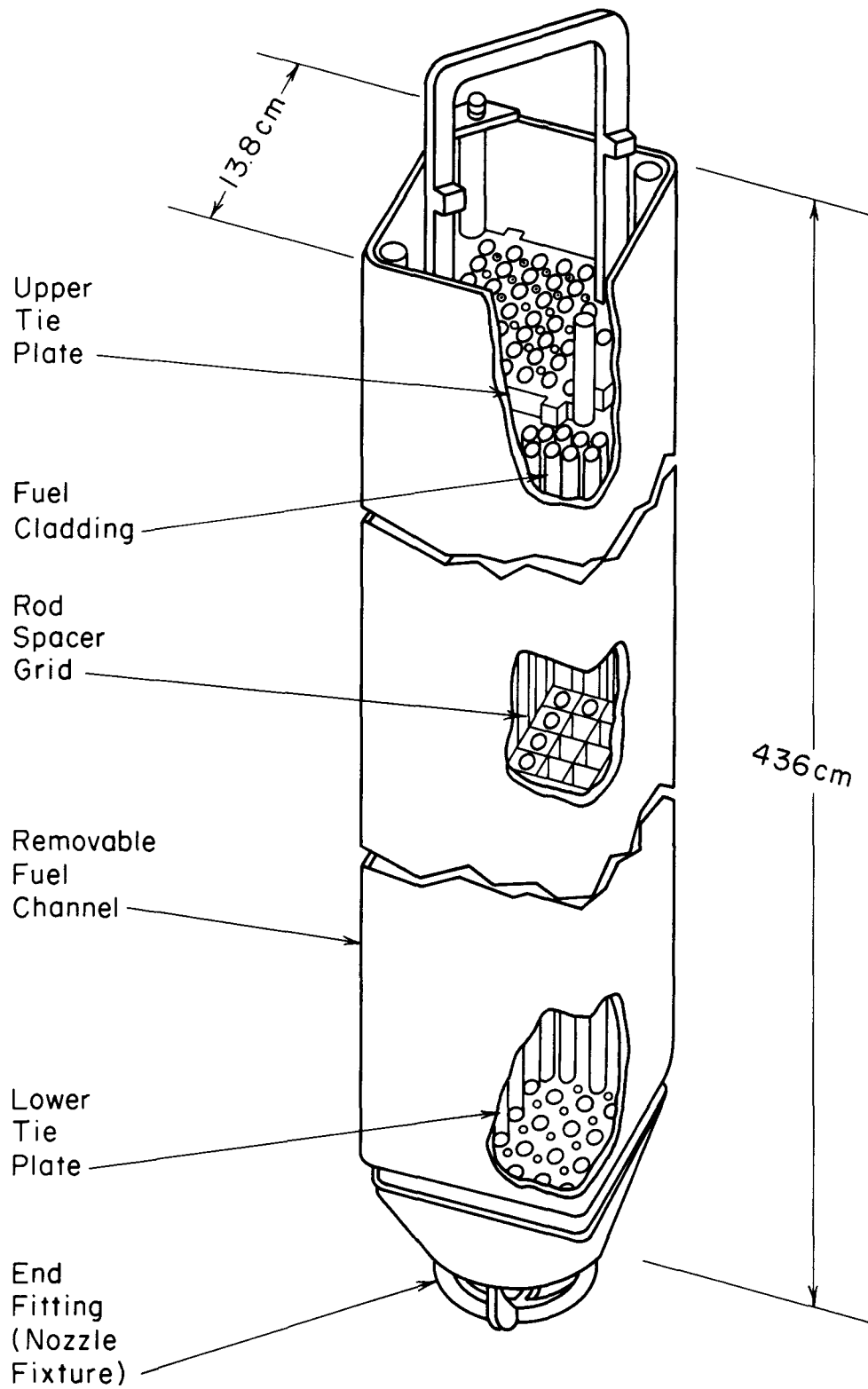


Fig. 4 Boiling Water Reactor Fuel Rod Assembly [27].

Table 3. Design Properties of Fuel Rod Assemblies

	Older PWR	Newer PWR	Older BWR	Newer BWR
Rod array	15 x 15	17 x 17	7 x 7	8 x 8
Cross section (cm)	21.4 x 21.4	-----	113.8 x 13.8	-----
Number of fuel rods per assembly	208	264	49	63
Number of non-fuel rods per assembly	17	25	0	1
Overall assembly length (cm)	405.9	-----	434.7	-----
Active fuel height (cm)	366	366	366	376
Rod center-to-center pitch (cm)	1.42	1.26	1.87	1.62
Fuel rod outside diameter (cm)	1.07	0.95	1.43	1.25
Clad thickness (cm)	0.066	0.058	0.082	0.086
Channel thickness (cm)	-----	-----	0.203	0.305
Metric tons uranium per assembly (MTU)	0.456	0.461	0.195	0.189
Number of assemblies per core, typical reactors	117	193	764	732

Table 4. Heat Generated by Spent Fuel.

Years After Discharge	PWR (W/MTHM)	BWR (W/MTHM)
0	0.1742E+07	0.1204E+07
1	0.1043E+05	0.7787E+04
2	0.5640E+04	0.4307E+04
3	0.3498E+04	0.2807E+04
4	0.2568E+04	0.2045E+04
5	0.2011E+04	0.1629E+04
6	0.1688E+04	0.1384E+04
7	0.1387E+04	0.1230E+04
8	0.1353E+04	0.1127E+04
9	0.1259E+04	0.1052E+04
10	0.1189E+04	0.9963E+03
20	0.8904E+03	0.7464E+03
30	0.7394E+03	0.6199E+03
40	0.6246E+03	0.5253E+03
50	0.5340E+03	0.4509E+03
100	0.2886E+03	0.2490E+03
200	0.1627E+03	0.1437E+03
400	0.1099E+03	0.9799E+02
600	0.8455E+02	0.7576E+02
1,000	0.5547E+02	0.5018E+02
10,000	0.1389E+02	0.1286E+02
30,000	0.5507E+01	0.5061E+01
100,000	0.1069E+01	0.9469E+00
1,000,000	0.3919E+00	0.3463E+00

Note:	<u>PWR</u>	<u>BWR</u>
Average Specific Power MW(t)/Metric Ton	37.5	25.9
Average Fuel Burnup MWd/Metric Ton	33,000	27,500

and heat transfer correlations for a fuel rod assembly of higher specific generation rate will provide bounding values for heat transfer coefficients and maximum temperature differences.

There are two inhibiting factors as far as exact modeling of a PWR fuel rod assembly is concerned. The first is the height of the assembly (4.06 m), and the second is the number of rods (204 or 208 active rods arranged in a 15 x 15 array in typical assembly). An active fuel height of 3.66 m (12 ft), a rod diameter of 1.07 cm (0.42 in.), and a complex internal construction (Fig. 5) add to the difficulties of constructing an exact replica in the laboratory. The length of the rods is a factor owing to space limitations in the laboratory. Thus, a motivation exists for constructing a thermally equivalent rod bundle in order to reduce the number of rods needed for the simulation of the typical PWR fuel rod assembly.

To decrease the number of heated rods in a laboratory experiment, the thermal characteristics of a typical rod have been determined with the following considerations. A typical three-year old PWR spent fuel assembly has a thermal output of about 2,200 W, based on a burn up of 38,000 MWd/MTM in a 204-rod assembly. Thus, average power output per rod is 10.78 W; and with an active fuel length of 3.66 m, the specific power output is approximately 2.94 W/m. The assembly can be divided into twenty-five subassemblies, each containing nine rods (see Fig. 6). For subassemblies containing nine active fuel rods, power output per meter is 26.46 W/m, while for subassemblies containing eight active rods, power output is 23.52 W/m. The area of the subassembly is 18.4 cm² (4.29 cm x 4.29 cm).

The length of the heated rods in the experiments is smaller than that in an actual assembly. For example, a 1.83 m (6 ft) active length with the thermal power averaged over the entire length, would result in a model fuel rod which produces approximately 50% of the theoretical normalized power. The 1.83 m active length permits a large height-to-diameter ratio to be maintained and the specific power

production allows a large range of Grashof numbers. Under these conditions, end effects are minimized with moderate total temperature differences between the heated rods and canister wall.

3.3 Experimental Apparatus

The major design components of the experimental apparatus include the cylinder, end plates, heater rods, cooling jacket, pressure control, and power supply with associated circuitry. Figure 7 presents a schematic of the apparatus and instrumentation. In the following paragraphs, detailed descriptions of the several components of the apparatus are presented.

The end plates on the cylinder provide both a gas tight seal and a means for introducing and mounting the heater rods into the cavity. Two designs of the top plate have been specified, one for the "thermally equivalent rod" and the other for the subassembly with nine heater rods in a 3 x 3 square array.

The top plate for the thermal equivalent rod is a 2.54 cm (1 in.) thick piece of aluminum with a diameter of 12.7 cm (5 in.). A shoulder of 1.27 cm (0.5 in.) width and a depth of 0.54 cm (0.21 in.) is cut into the plate and threaded for fastening with the cylinder, which is also threaded internally. At the center of the plate is a 1.91 cm DIA (0.75 in. DIA) hole which is threaded to accept the fitting at the tip end of the heater rod. Two 0.32 cm DIA (0.13 in. DIA) holes on one-inch centers are drilled into the top plate for inserting thermocouples. For the purpose of introducing either air or another gas into the cylinder, a 0.64 cm DIA (0.25 in. DIA) threaded hole on a 2.87 cm (1.13 in.) center is drilled into the top plate. A groove with a width of 0.36 cm (0.14 in.) and depth of 0.19 cm (0.08 in.) with an inside diameter of 8.89 cm (3.50 in.) is cut into the lower surface of the shoulder. This groove is needed to accommodate an O-ring for the purpose of sealing the cylinder. Two 0.68 cm DIA (0.26 in. DIA) holes on 5.69 cm (2.24 in.) centers with a depth of 0.71 cm (0.28 in.) are drilled into the top

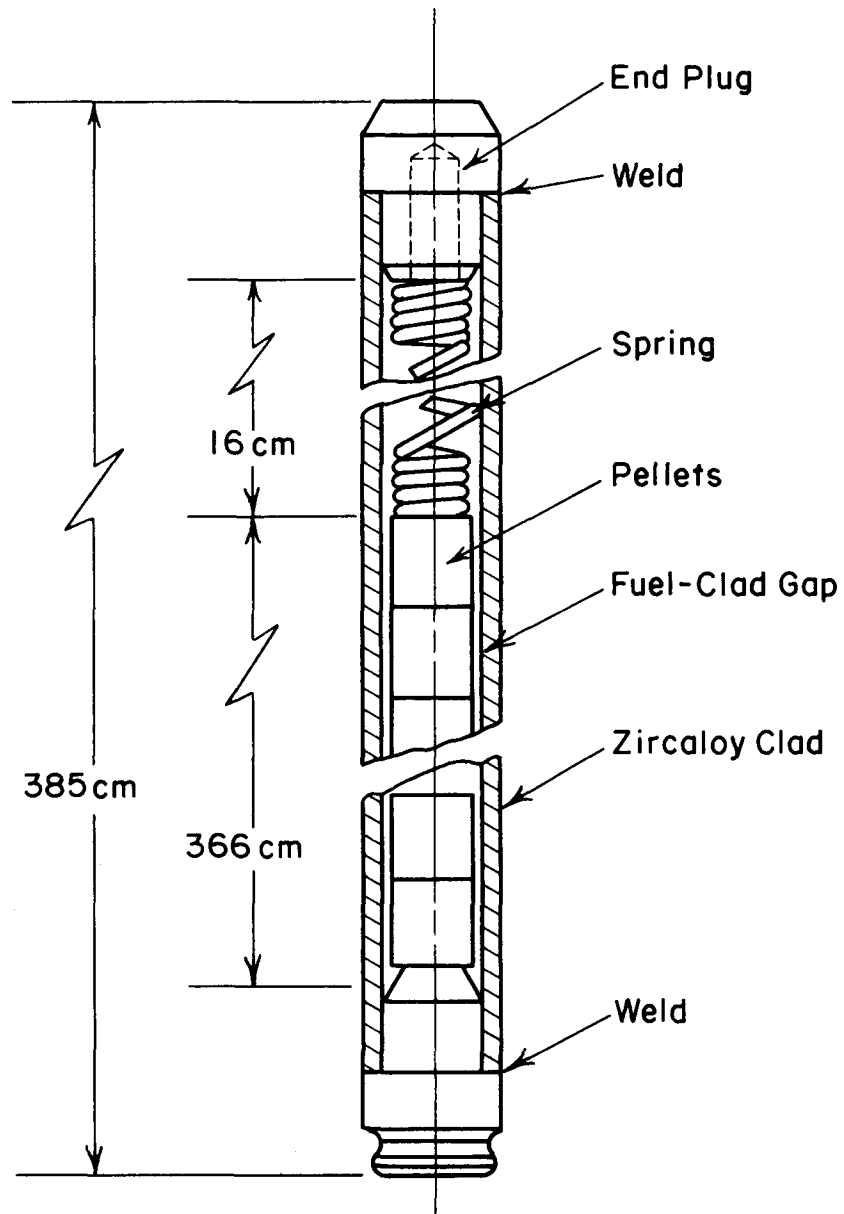


Fig. 5 Typical Pressurized Water Reactor Fuel Rod [28].

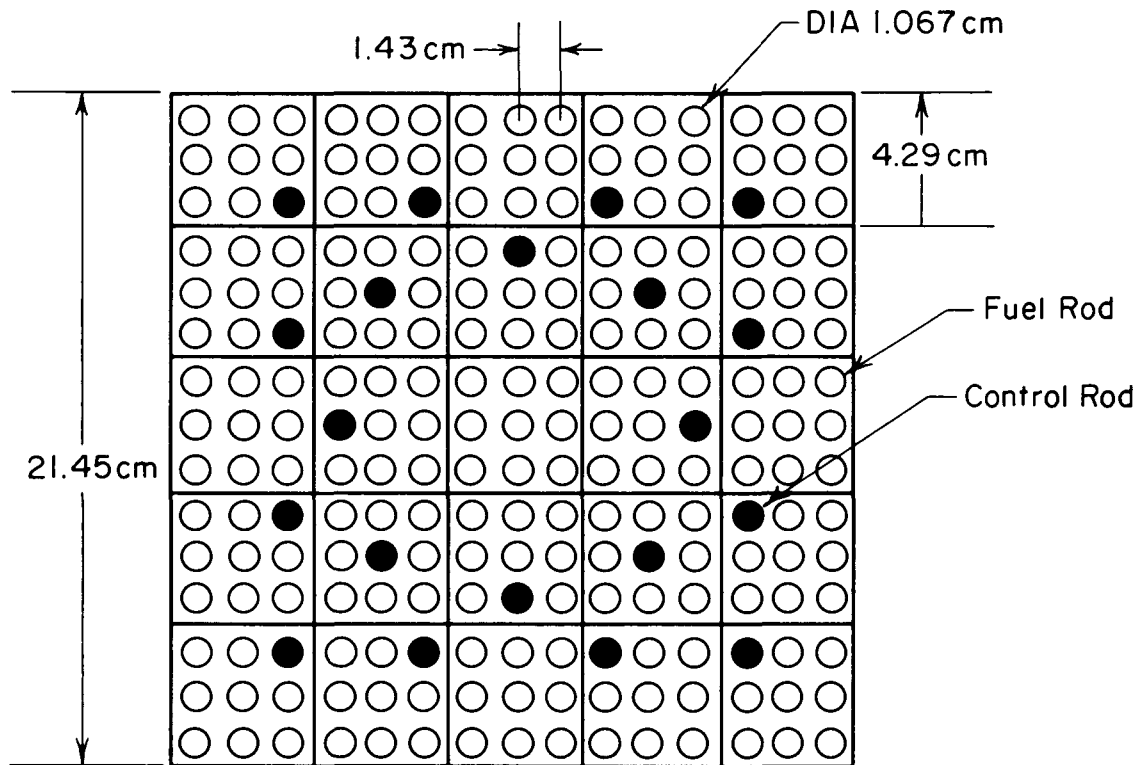


Figure 6 Cross Section of a Typical Pressurized Water Reactor Fuel Rod Assembly [28].

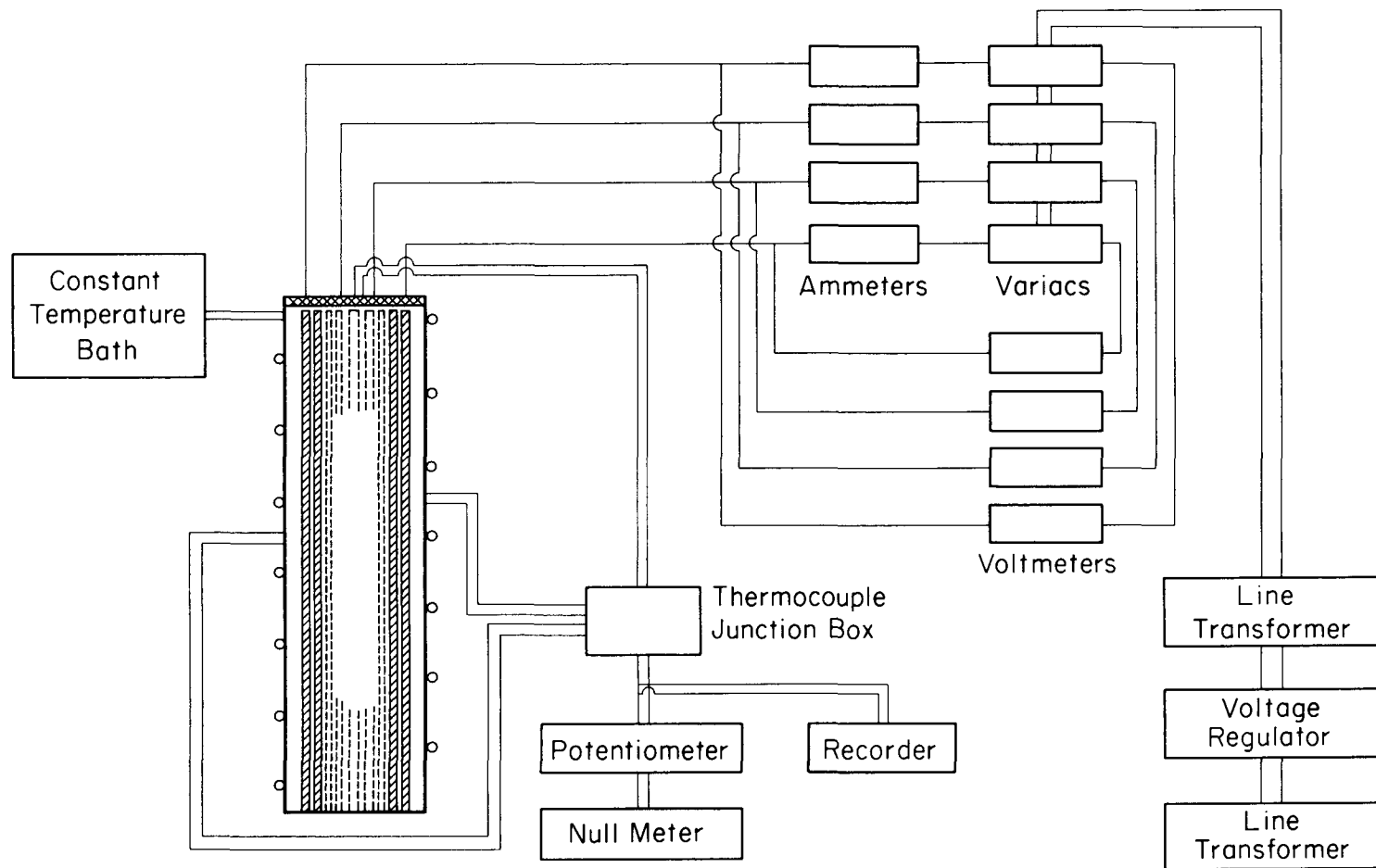


Fig. 7 Schematic of Experimental Apparatus.

surface of the plate. These holes facilitate the tightening of the top plate into the cylinder with the aid of a spanner wrench.

In the case of the nine-rod bundle on a 3 x 3 square array, the top plate is a 2.86 cm (1.13 in.) thick piece of aluminum with a diameter of 12.70 cm (5 in.). The upper surface is counterbored 0.32 cm (0.13 in.) with a diameter of 9.60 cm (3.78 in.). Nine holes of 0.64 cm DIA (0.25 in. DIA) are drilled and tapped to accept the fittings at the top ends of the rods. These holes are on a square grid 3.92 cm x 3.92 cm (1.54 in. x 1.54 in.), which fits inside a circle of 8.26 cm DIA (3.25 in. DIA). The pitch distance between the holes is 1.95 cm (0.77 in.). Nine holes of 0.32 cm DIA (0.13 in. DIA) are also made in convenient locations in the top plate for insertion of thermocouples into the cylinder. A flange 1.59 cm (0.63 in.) thick (measured from the top surface of the plate) is made by cutting a shoulder of 1.27 cm (0.50 in.) thick (measured from the bottom surface of the plate) and a depth of 0.54 cm (0.21 in.) into the plate. Due to the space limitations, the gas inlet is located on the side of the 1.59 cm (0.63 in.) thick flange of the plate. A 0.64 cm DIA (0.25 in. DIA) threaded hole is drilled in the side of the flange of the top plate. The hole is 1.1 cm (0.44 in.) deep, thereafter the diameter is reduced to 0.32 cm (0.13 in.) with a length of 1.77 cm (0.69 in.). A hole of 0.32 cm (0.13 in.) is drilled from the bottom face of the plate on 3.46 cm (1.36 in.) centers to meet the inlet air flow hole. The holes to facilitate the use of the spanner wrench and the groove for the O-ring are specified in the same manner as for the top plate in the case of a thermally equivalent rod.

In order to assure that the 91.44 cm (36 in.) long nine-rod bundle on a 3 x 3 square array will maintain the desired pitch distance, a guide is provided. The guide is a 0.32 cm (0.13 in.) thick piece of aluminum with a diameter of 8.23 cm (3.23 in.). Nine holes of 0.64 cm DIA (0.25 in. DIA) are drilled and tapped on the side of the guide in line with the major diagonal of the array. The guide can be attached to the bottom of the square array of the heater elements with the aid of

two screws through the two side holes in the guide. Figure 8 contains a view of the top plate for the square array and the guide.

The bottom plate is basically the same as for the case of the single rod, with the exception that it has no holes in it.

The cylinder is 8.26 cm ID x 9.53 cm (3.25 in. ID x 3.75 in. OD) and 92.17 cm (36.5 in.) long. It is made of carbon steel. Figure 10 presents the details of the cylinder. For the purpose of maintaining constant temperatures on the surface of the cylinder, 30.48 m (100 ft) of 0.9 cm DIA (0.38 in. DIA) soft copper tubing is used to provide a cooling coil around the cylinder. This coil is wound in a counter flow manner to give a uniform surface temperature. A gap of 0.64 cm (0.25 in.) is provided between the consecutive turns of cooling coil. This gap is needed for emplacement of the thermocouples along the cylinder. Figure 10 contains a view of the cylinder with the cooling coil in place.

Thirty-two holes 0.32 cm (0.13 in.) deep are drilled along the length of the cylinder. These holes are along four lines (eight holes per line) that divide the cylinder to four quarters. The temperature difference across a cylinder with inside diameter of 8.26 cm (3.25 in.) and thickness of 0.32 cm (0.13 in.) for 1 W of heat transfer is calculated for one-dimensional heat conduction. For every watt of cooling load, the thermocouple output will be $3 \times 10^{-4}^{\circ}\text{C}$ below the actual inside surface temperature. The maximum cooling load for these experiments will be about 120 W, which will result in 0.04°C difference between thermocouple readings and the actual surface temperature.

Another set of calculations have been performed in order to check the validity of the assumption that the heat transfer will only take place in the radial direction. The ratio of thermal resistance in the axial direction to that in the radial direction is on the order of 10^6 . This indicates that the assumption of constant temperature along the cylinder is a valid one.

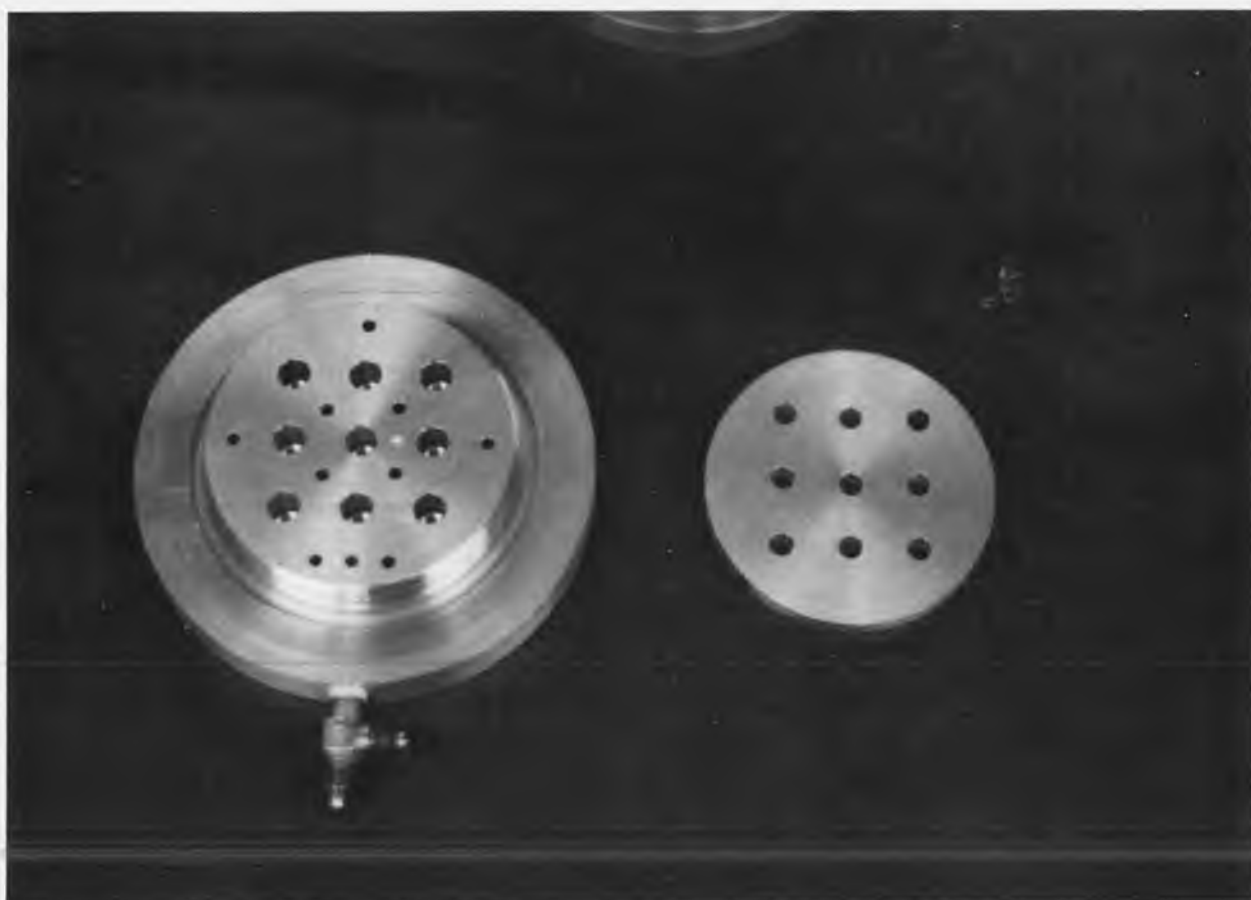


Fig. 8 View of the Top Plate for the Square Array with Guide.

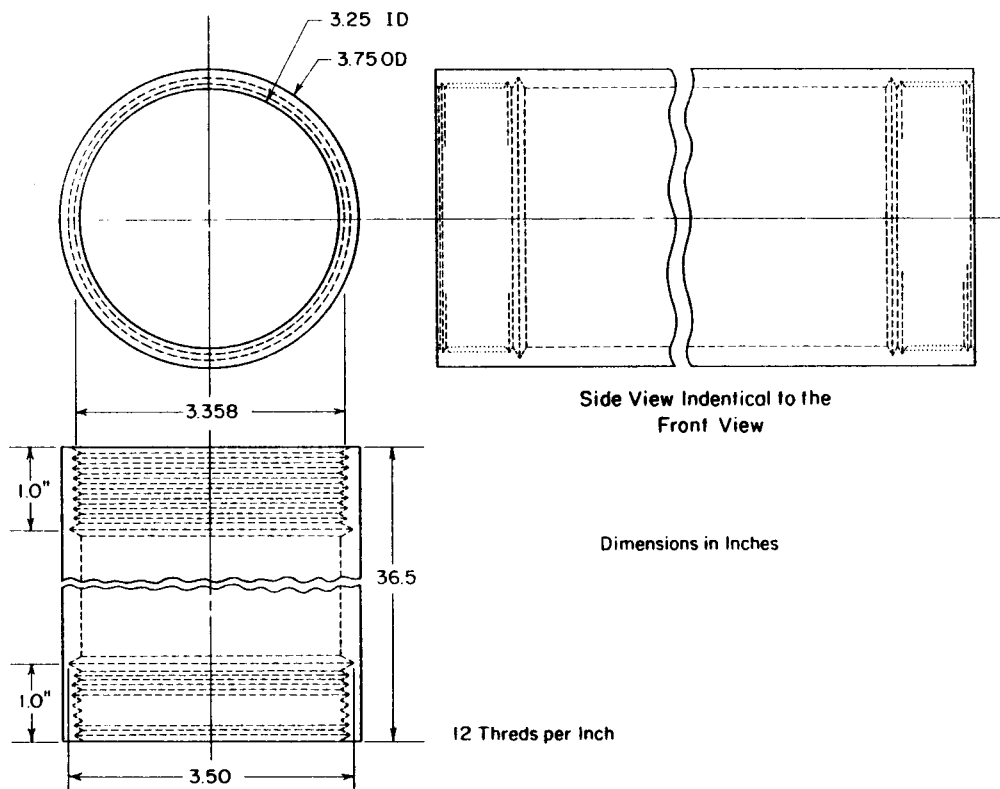


Fig. 9 Detail of Cylinder

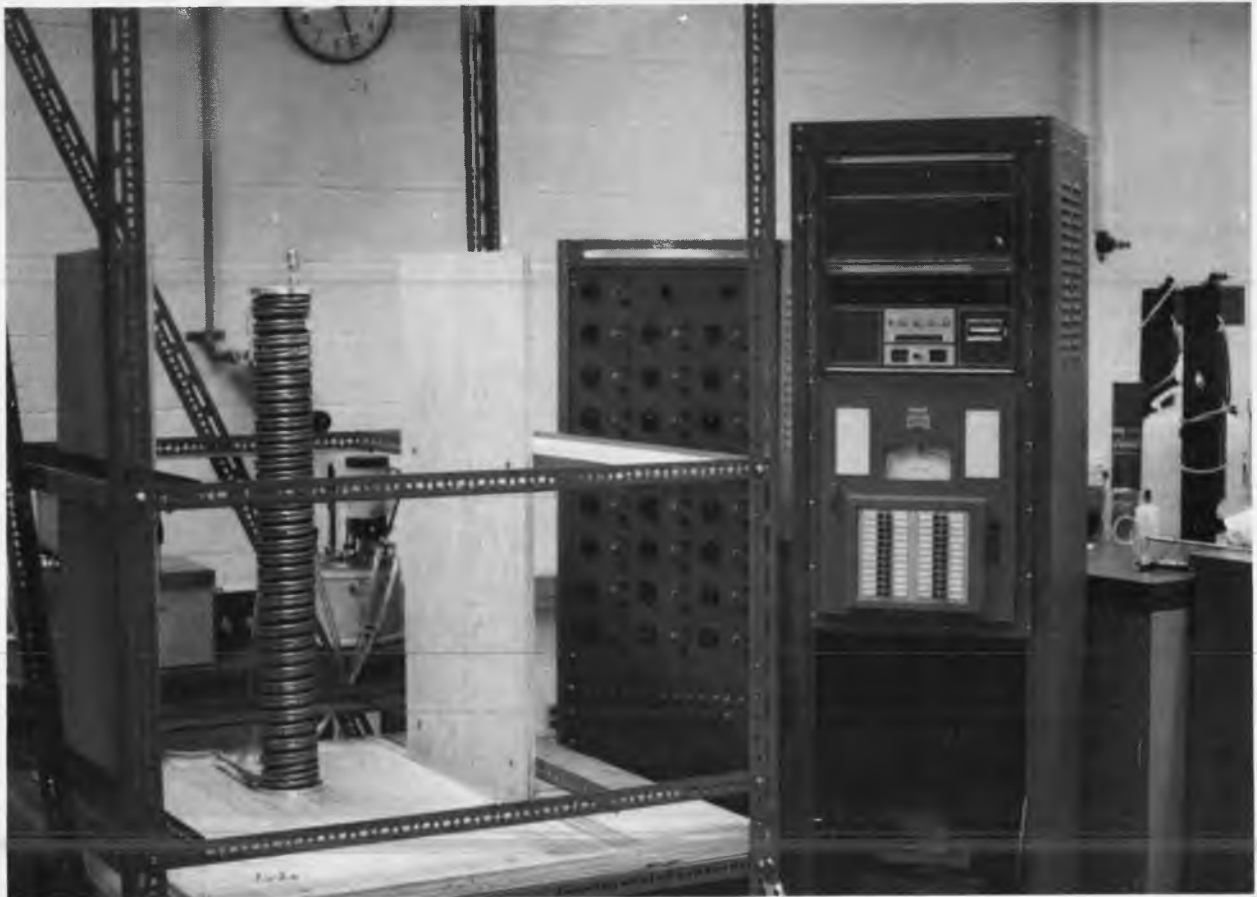


Fig. 10 View of Experimental Apparatus, Stand, Power Control Panel and Data Logger.

Heater elements for the 3 x 3 rod array are 0.635 cm DIA (0.25 in. DIA) and 91.44 cm (3 ft) long. This length includes the threaded fitting at the top of the heater element which is 1.27 cm (0.5 in.) long. The heater element for the thermally equivalent rod is the same length but 1.91 cm CIA (0.75 in. DIA). The threaded fitting in this case is 2.54 cm (1.0 in.) long. In either case, the heater elements are covered with a stainless steel sheath. At the top of the elements is a 2.54 cm (1 in.) long unheated region, including the threaded fitting. The fitting permits easy sealing of the heater rod to the top plate. The power rating of both types of heaters is 3.937 W/cm (120 W/ft). This power density provides sufficient heating to approximate the power output of three-year old spent fuel from a PWR with 38,000 MWd/MTHM burnup. The active heated length on the rods is 87.63 cm (34.5 in.).

Surface temperatures of the heater elements are monitored with thermocouples inserted through the 0.32 cm DIA (0.13 in DIA) holes in the top plate. Copper-constantan thermocouples are used, and their junctions are pressed against the surface of the heater element with a high temperature tape. Temperature data are recorded with a high sensitivity voltmeter and a multi-channel data logger (see Fig. 10).

The cylinder and cooling coil are covered by Styrofoam insulation. The insulation assembly is 60.96 cm x 60.96 cm and is shown in Fig. 11.



Fig. 11 View of Apparatus with Thermal
Insulation Installed.

(c)



SECTION 4. RESULTS

Heat transfer results obtained in the present study include temperature distributions on the cylinder wall and outer surface of the heater rod, as well as heat transfer coefficients for free convection and combined free convection and thermal radiation for Rayleigh numbers from 200 to 2×10^6 . This range of Rayleigh numbers spans the conduction to turbulent regimes of convection as defined in the study of Thomas and de Vahl Davis [23]. In all cases, a correction for thermal radiation in the annulus is applied in the calculation of the coefficient for free convection heat transfer. This correction includes the effects of surface emissivities and radiative view factors.

Experimentally, the correction for thermal radiation was obtained by operating the apparatus at 0.02 atm pressure over a range of power inputs. An overall conductance for thermal radiation was defined as

$$C_r = \frac{q - q_{\text{convection}}}{\sigma(T_i^4 - T_o^4)_{\text{vacuum}}}, \quad (12)$$

where

$$C_r = \frac{1}{\frac{1}{\epsilon_i} - 1 + \frac{1}{F_{i-o}} + \frac{A_i}{A_o} \left(\frac{1}{\epsilon_o} - 1 \right)} \quad (13)$$

Values of C_r as a function of mean temperature of the test gas, $(T_{mR} + T_{mC})/2$, were then used to correct for thermal radiation in experiments at atmospheric and higher pressures.

In Eq. (12), the quantity $q_{\text{convection}}$ is the heat transfer across the annulus by convection at the "vacuum condition" of 0.02 atm. This convective heat transfer rate was obtained for the Rayleigh numbers of the vacuum runs ($Ra \leq 80$) by use of the Thomas and de Vahl Davis correlation which compares favorably with the present work in the conduction regime.

Values of C_r as a function of mean temperature of the test gas are presented in Fig. 12. A simple equation of the form

$$C_r = 0.946 - (1.0946 \times 10^{-3})T_f \quad (14)$$

adequately describes the data for $25^\circ\text{C} \leq T_f \leq 110^\circ\text{C}$. Experimental data for the runs in which C_r was determined are given in Appendix A.

It should be noted that Eq. (14) holds only for the present apparatus and for the span of time that the present experiments were conducted. Depending on the extent of oxidation that takes place on the surfaces of both the heater elements and the cylinder, Eq. (14) would have to be determined again for future experiments. Nevertheless, Eq. (14) provides a useful means to quickly estimate the radiative contribution to the overall heat transfer in the above mentioned temperature range. To lessen the possibility of oxidation effects on C_r , the apparatus was filled with helium between runs and kept at temperatures slightly above ambient to minimize condensation.

Once the overall conductance for thermal radiation was known, it was used to determine the equivalent heat transfer coefficient for thermal radiation at a given power level. The usual linearization of the radiation transport between the surfaces of the annulus gives,

$$h_r = C_r \sigma (T_i^2 + T_o^2)(T_i + T_o). \quad (15)$$

The temperatures used to find h_r were taken from measurements at atmospheric pressure in a given run. Also, C_r was evaluated at mean temperature of the gas of that same run. The radiation heat transfer coefficient was subtracted from the total heat transfer coefficient, which is the ratio of the total heat flux to the temperature difference

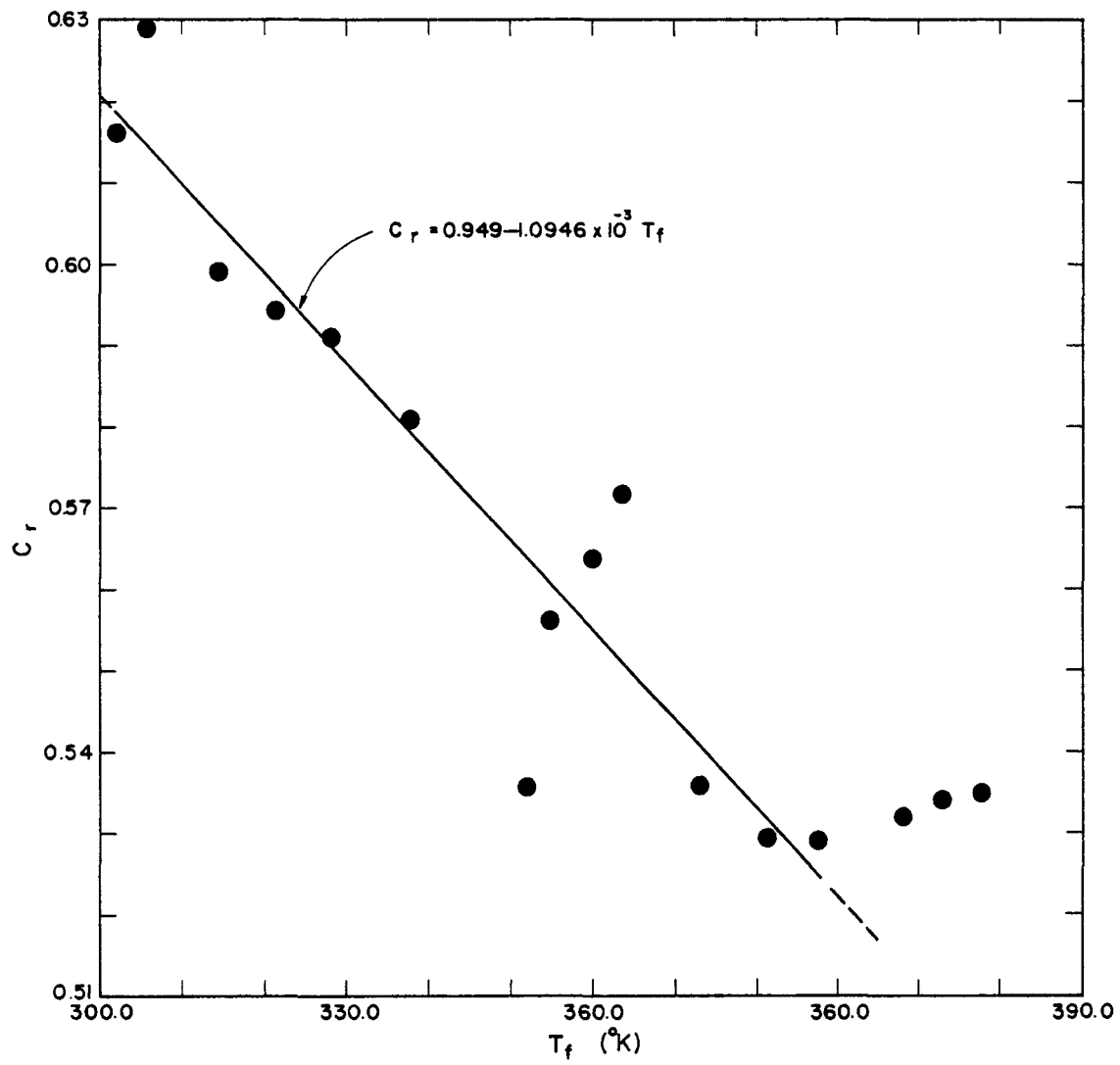


Figure 12. Overall Thermal Conductance for Radiative Transport in the Annulus.

across the annulus. The resulting quantity is the corrected coefficient for free convection alone. Experimental data are tabulated in Appendix A, Tables A6 to A9.

Figures 13 and 14 contain temperature distributions along the rod with power input as a parameter. The data of Fig. 13 were obtained at 0.02 atm, while the data of Fig. 14 were obtained at atmospheric pressure. The test gas was air for these data. It is evident that a reduction in pressure results in a sharp increase in temperature. It should also be noted that the temperature measurement at approximately the midpoint of the rod is somewhat distorted due to a slight separation of the thermocouple from the rod surface.

Figures 15 to 17 contain temperature distributions on the rod with helium as the test gas. The data of Figure 15 were obtained at atmospheric pressure, while the pressure for the data of Figs. 16 and 17 were 2.0 atm and 3.0 atm respectively. In comparing Figs. 14 and 15, it is evident that, for approximately the same power input and the same pressure, the rod surface experiences lower temperatures with helium than with air as the test gas. This is due to the higher thermal conductivity of helium. Furthermore, Figs. 15 to 17 indicate that an increase in the pressure, for approximately the same power input, would result in a decrease in the rod surface temperatures. This behavior is expected, since a doubling of density results in a four fold increase in the Rayleigh number.

The temperature on the cylinder is held constant to an acceptable level by the cooling water circulating through the copper coil attached to its surface. Figure 18 contains a typical temperature distribution on the surface of the cylinder. The mean temperature for this particular distribution is 24.8°C with a standard deviation of 0.3°C.

The temperature difference across the annulus as a function of the power is presented in Fig. 19. These data show the generally non-linear relation between $T_{mR} - T_{mC}$ and rod power, which results because of the radiative transport.

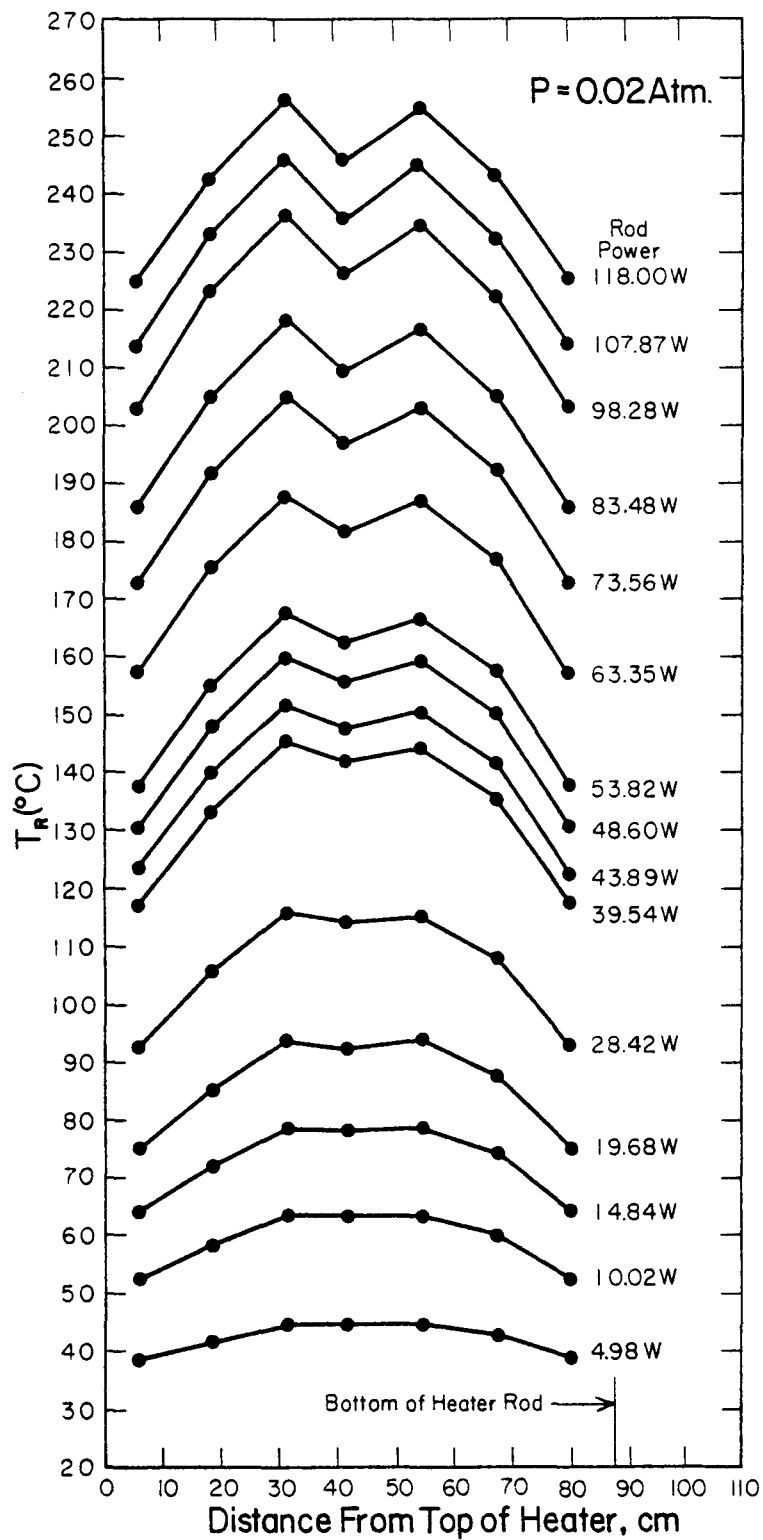


Figure 13. Temperature Distribution on Rod Surface at 0.02 Atm. Test Gas is Air.

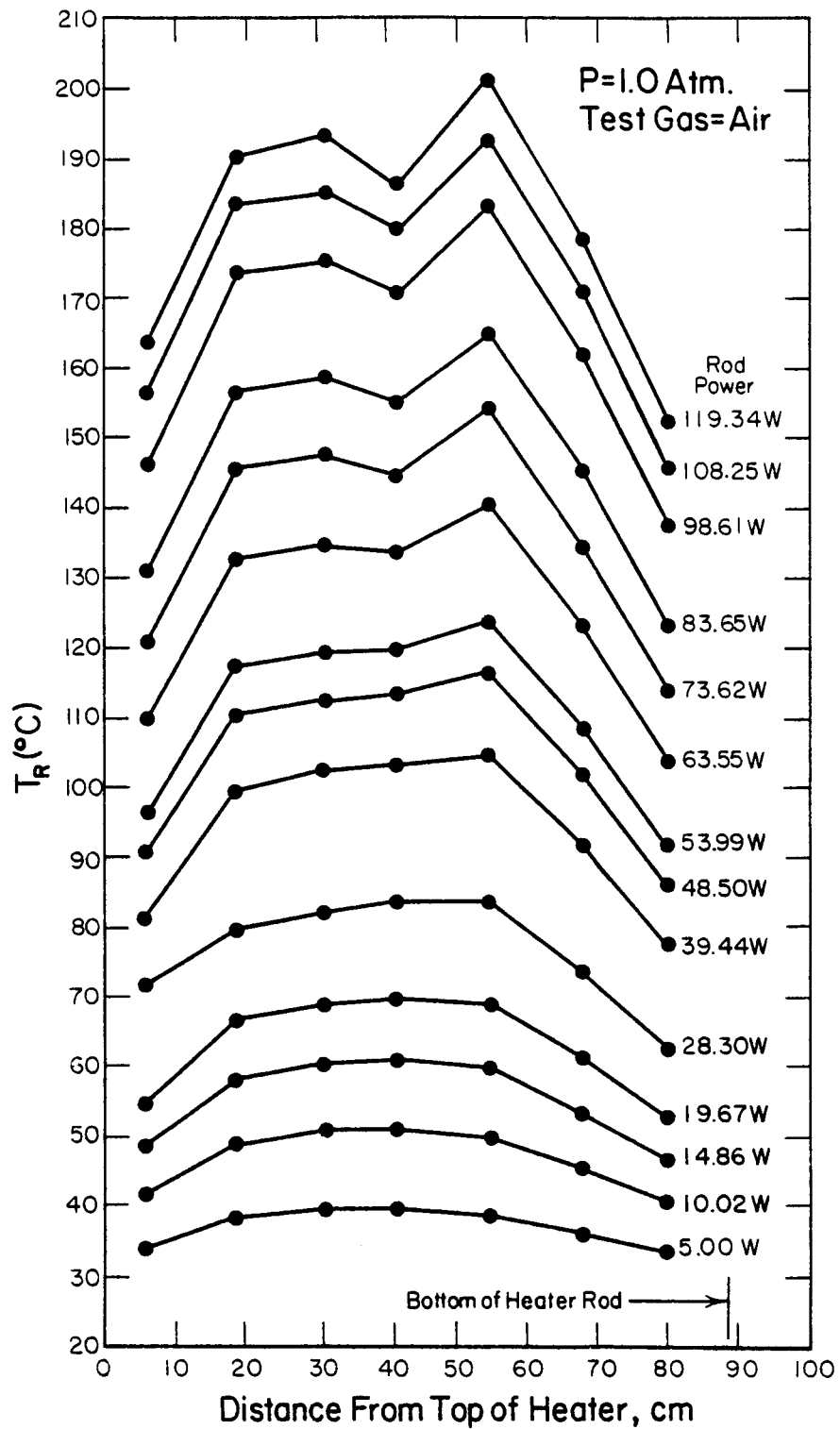


Figure 14. Temperature Distribution on Rod Surface at Atmospheric Pressure. Test Gas is Air.

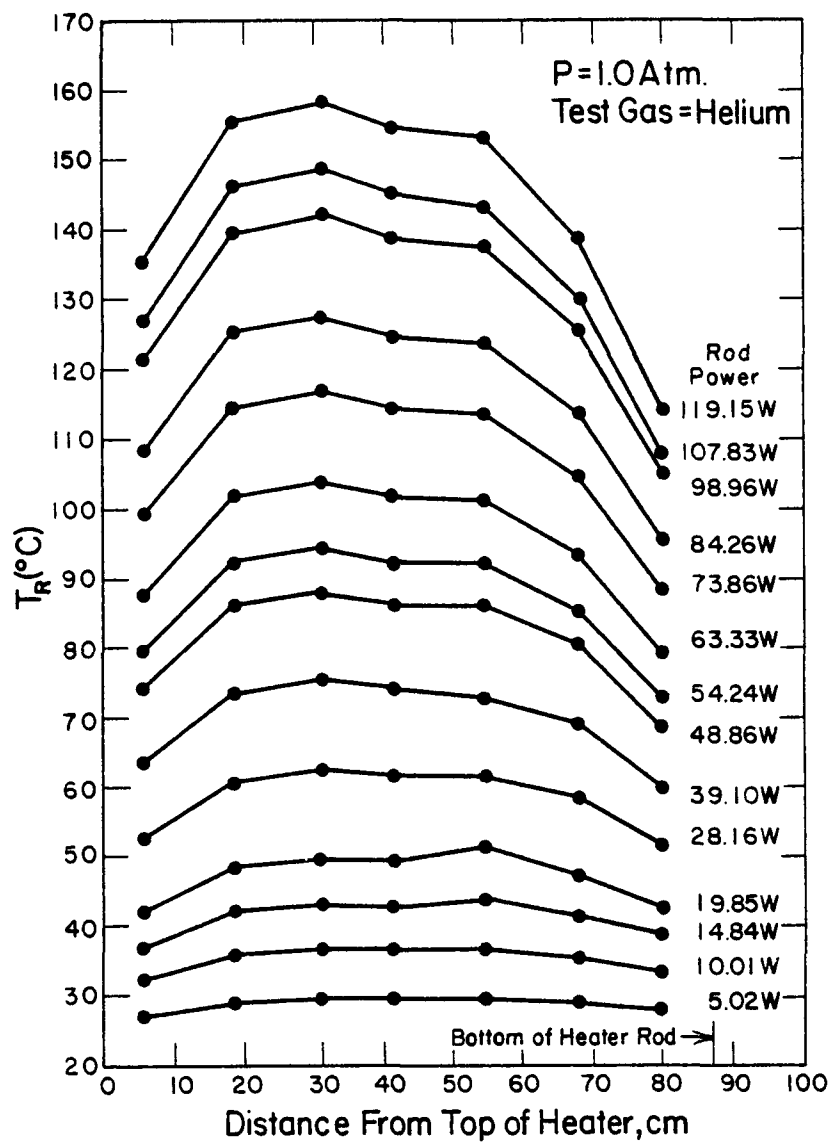


Figure 15. Temperature Distribution on Rod Surface at Atmospheric Pressure. Test Gas is Helium.

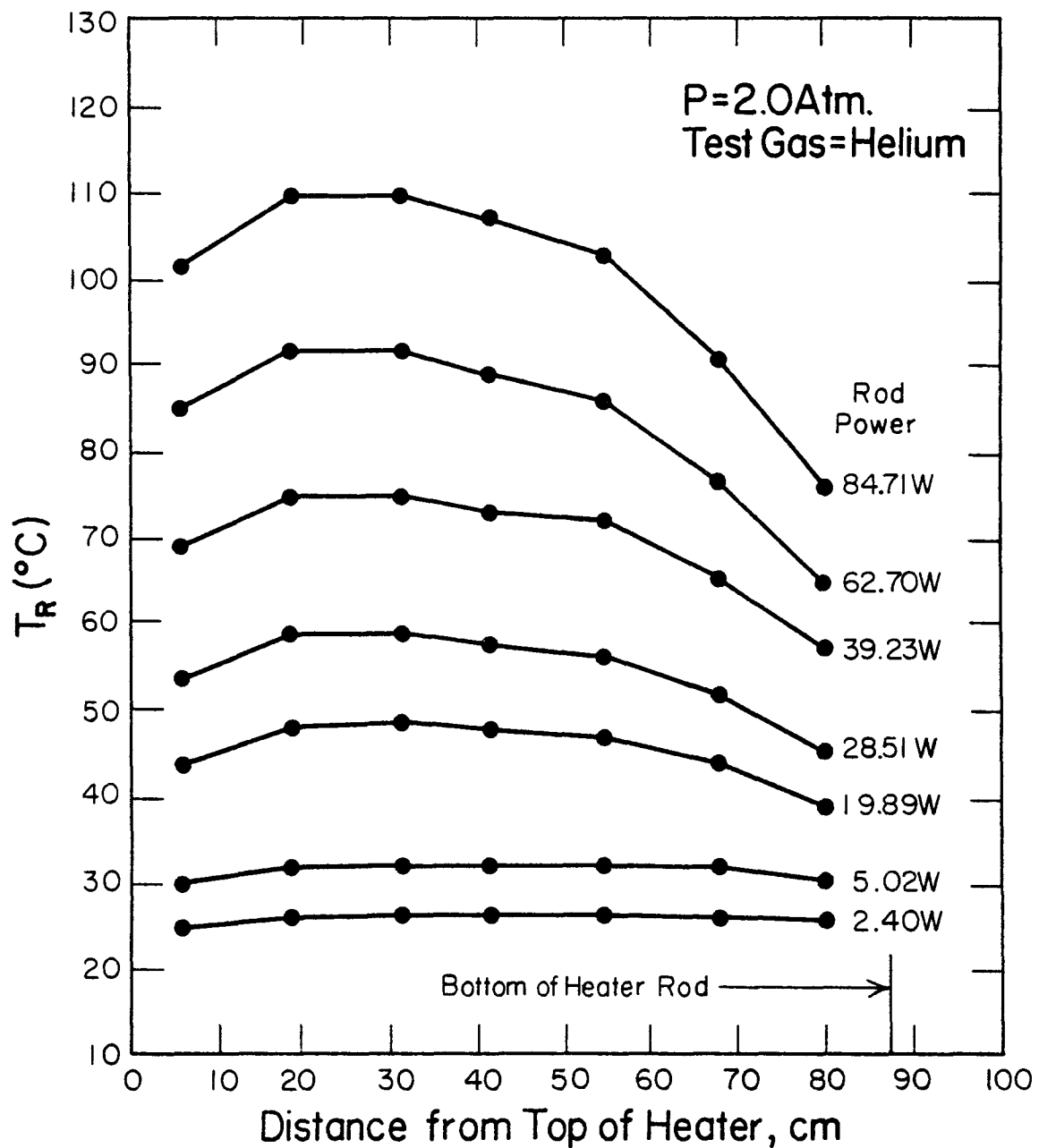


Figure 16. Temperature Distribution on Rod Surface at 2.0 Atm. Test Gas is Helium.

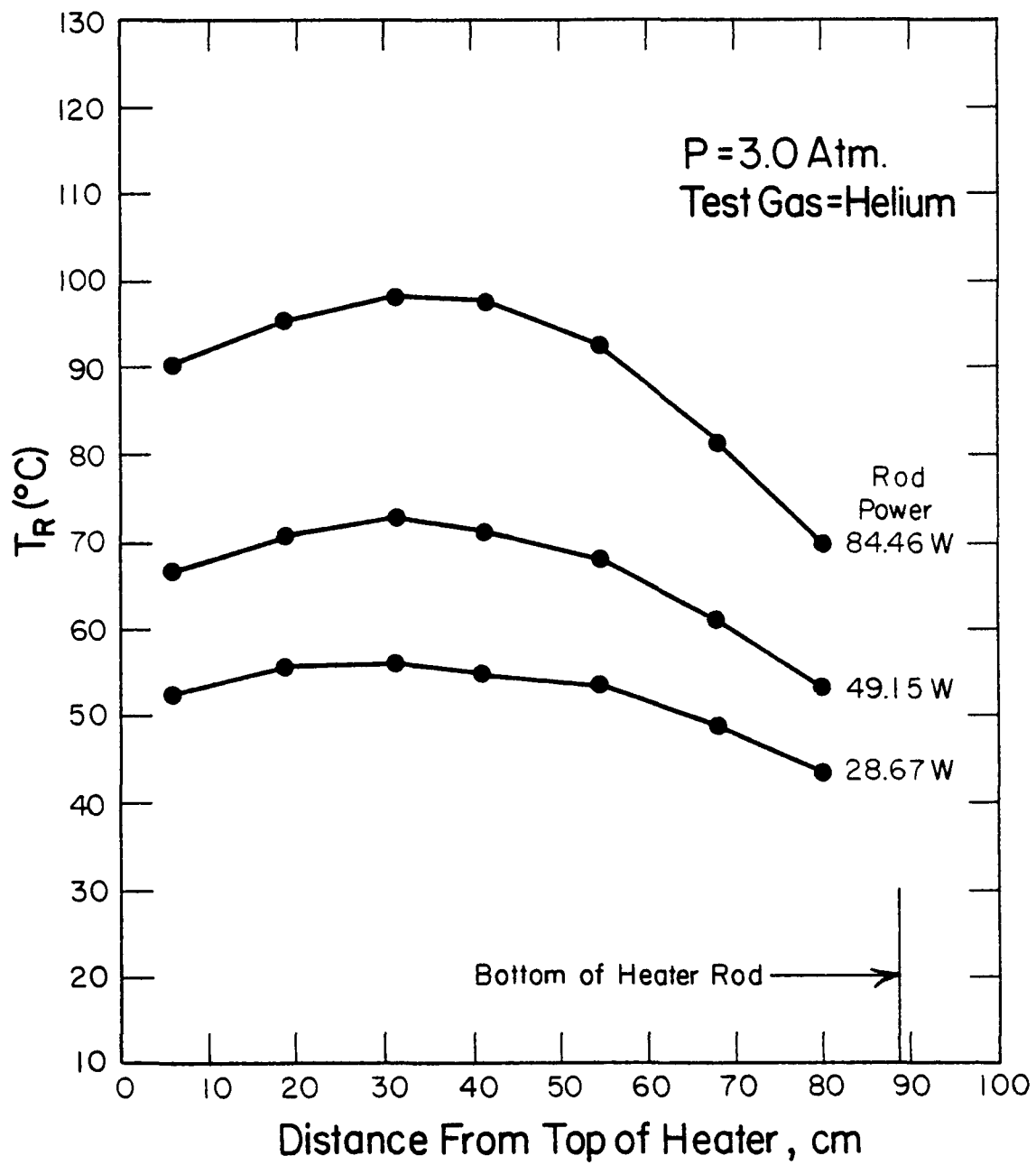


Figure 17. Temperature Distribution on Rod Surface at 3.0 Atm. Test Gas is Helium.

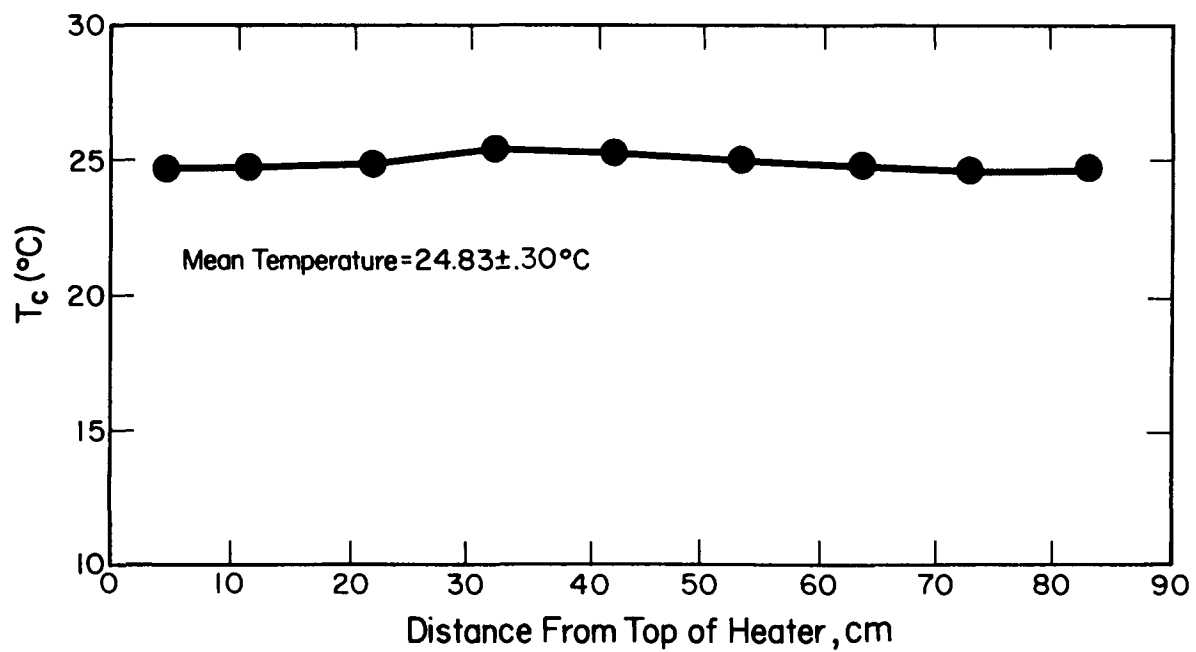


Figure 18. Surface Temperature of Cylinder as a Function of Distance from Top of Heater.

Experimental Nusselt numbers are presented in Fig. 20. Based on the work of Thomas and de Vahl Davis [23], the present study covers the conduction, transition and a portion of the boundary layer, or turbulent regimes. Generally, present value are somewhat higher than the results of Thomas and de Vahl Davis above the conduction regime. This difference is a result of the thermal boundary condition on the inner cylinder and the limitation of the numerical computation of Thomas and de Vahl Davis to strictly laminar flow. A linear regression of $\ln(\text{Nu})$ on $\ln(\text{Ra})$ gives the following correlation:

$$\text{Nu} = 1.406 \text{ Ra}^{0.077} \quad (16)$$

for $1 \times 10^3 \leq \text{Ra} \leq 6 \times 10^3$,

and

$$\text{Nu} = 0.163 \text{ Ra}^{0.322} \quad (17)$$

for $6 \times 10^3 < \text{Ra} \leq 2.33 \times 10^6$.

Heat transfer across the annulus by thermal radiation, as a fraction of total heat transfer, is given by the heat transfer coefficient for radiation defined in Eq. (15). Generally, heat transfer by radiation varies with the Rayleigh number and test gas. With air as the test gas, thermal radiation can account for up to 50% of the total heat transfer. With helium, radiation can account for up to one-third of the total. Values of the fraction of energy transport by convection alone are given in Appendix A, Tables A6 to A9.

An analysis of the experimental uncertainty in the present data is presented in Appendix B. By summing the total fractional uncertainty in the measured quantities and thermophysical properties, one obtains a 6% to 8% uncertainty in the Rayleigh number and an 8% to 10% uncertainty in the Nusselt number.

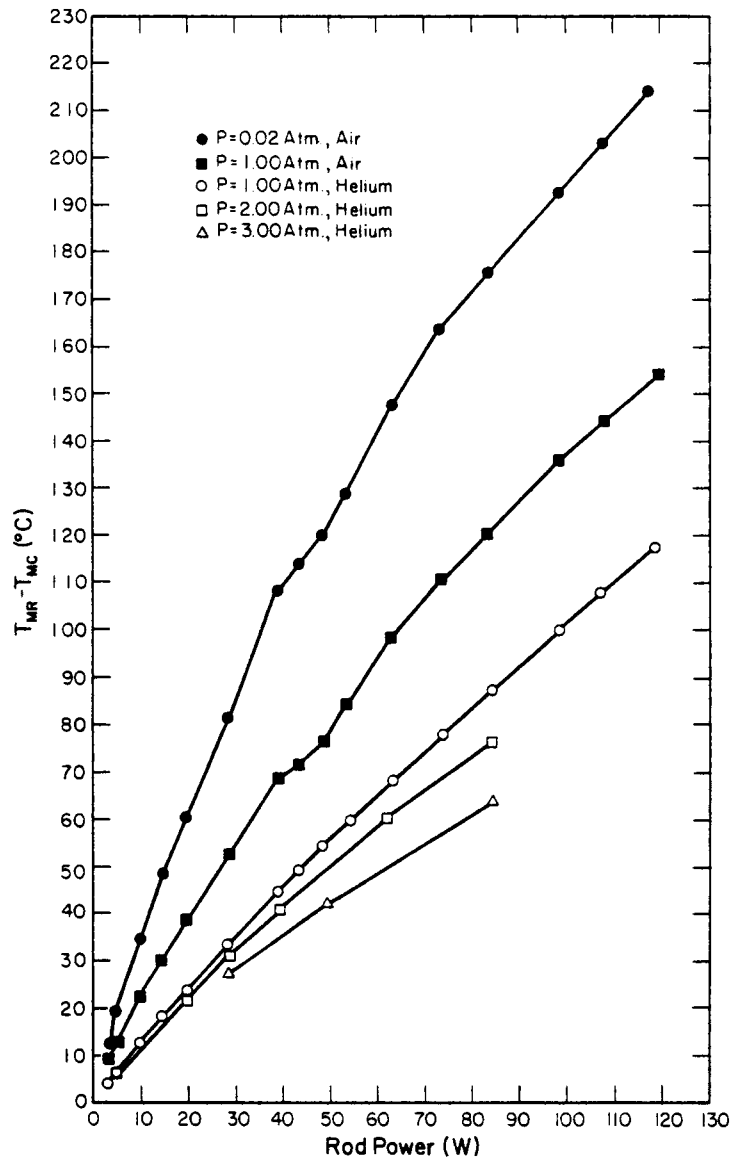


Figure 19. Temperature Difference Across the Annulus as a Function of Rod Power.

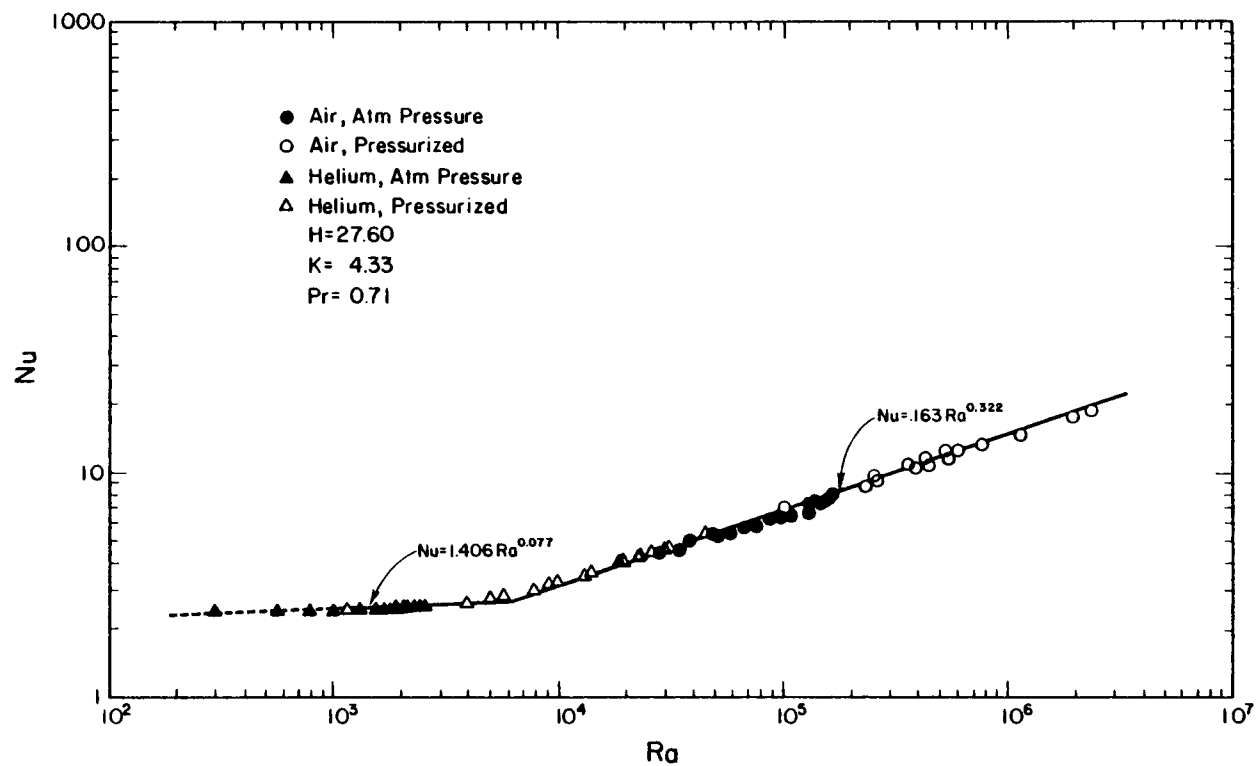


Figure 20. Nusselt Number as a Function of Rayleigh Number. Values are Corrected for Radiation Effects.



APPENDIX A. EXPERIMENTAL DATA



Table A1. Mean Temperature of Rod and Cylinder
Surfaces at P = 0.02 Atm.

Run No.	Rod Power (W)	Rod Temperature (°C)		Cylinder Temperature (°C)		T (°C)	T _f (°C)	C _r (Eq. (12))
		T _{mR}	Std. Dev.	T _{mC}	Std. Dev.			
14	2.45	34.03	1.65	24.63	0.15	9.4	29.33	0.616
8	4.98	42.3	2.6	24.34	0.09	17.96	33.32	0.629
4	10.02	58.94	4.95	24.43	0.14	34.51	41.68	0.599
6	14.82	72.81	6.5	24.49	0.21	48.32	48.65	0.594
10	19.68	85.74	7.95	24.91	0.21	60.83	55.32	0.591
2	28.42	106.19	9.94	24.33	0.27	81.86	65.26	0.581
12	39.54	133.38	11.93	24.86	0.44	108.52	79.12	0.536
16	43.89	139.44	12.01	25.43	0.5	114.01	82.43	0.556
20	48.6	147.71	12.18	27.55	0.53	120.16	87.63	0.564
22	53.82	154.88	12.47	26.65	0.57	128.23	90.76	0.572
24	63.35	174.48	12.89	27.04	0.71	147.44	100.76	0.536
26	73.56	190.2	13.2	27.08	0.84	163.12	108.64	0.529
28	83.48	203.51	13.42	27.72	0.95	175.79	115.61	0.529
30	98.28	221.1	13.61	28.54	1.12	192.56	124.82	0.532
32	107.87	231.47	13.29	28.53	1.24	202.94	130.0	0.534
34	118.00	241.83	12.76	28.20	1.36	213.63	135.02	0.535

Table A2. Mean Temperature of Rod and Cylinder
Surfaces at P = 1.0 Atm With Air as
Test Gas.

Run No.	Rod Power (W)	Rod Temperature (°C)		Cylinder Temperature (°C)		T (°C)	T _f (°C)	(Eq. (14))
		T _{mR}	Std. Dev.	T _{mC}	Std. Dev.			
70	4.00	35.29	1.99	25.19	0.06	10.10	30.24	0.617
7	5.00	36.87	2.64	24.37	0.11	12.50	30.62	0.617
71	6.00	39.03	2.73	24.84	0.09	14.19	31.93	0.615
74	8.05	40.46	3.1	21.76	0.12	18.70	31.11	0.616
72	8.07	42.99	3.34	24.87	0.13	18.13	33.93	0.613
3	10.02	46.78	4.49	24.44	0.15	22.34	35.61	0.611
90	14.5	60.69	4.49	30.99	0.21	29.69	45.84	0.60
5	14.86	55.39	5.89	24.52	0.24	30.87	39.96	0.606
9	19.67	63.17	7.08	24.56	0.30	38.61	43.86	0.602
1	28.3	76.76	7.90	24.61	0.39	52.15	50.69	0.595
91	29.32	83.71	6.94	31.18	0.42	52.53	57.44	0.587
11	39.44	94.25	11.20	25.19	0.53	69.06	59.72	0.585
15	43.82	96.97	11.40	25.19	0.58	71.78	61.08	0.583
19	48.50	104.42	11.98	27.54	0.63	76.88	65.98	0.578
21	53.99	110.83	12.78	26.48	0.69	84.35	68.66	0.575
23	63.55	125.44	14.03	27.00	0.81	98.44	76.22	0.567
25	73.62	137.28	14.97	27.17	0.93	110.11	82.23	0.56
27	83.65	147.77	15.60	27.68	1.04	120.09	87.73	0.554
29	98.61	163.89	16.73	28.59	1.22	135.30	96.24	0.545
31	108.25	173.43	17.04	29.14	1.34	144.24	101.31	0.539
33	119.34	180.85	17.30	27.74	1.45	153.11	104.3	0.536

Table A3. Mean Temperature of Rod and Cylinder
Surface at P = 1.5 Atm With Air as
Test Gas.

Run No.	P (Atm)	Rod	Rod		Cylinder		T (°C)	T _f (°C)	C _r (Eq. (14))
		Power	Temperature (°C)		Temperature (°C)				
		(W)	T _{mR}	Std. Dev.	T _{mC}	Std. Dev.			
73	1.5	8.05	37.87	2.57	21.74	0.13	16.13	29.8	0.618
82	1.5	28.88	68.91	5.41	22.83	0.41	46.08	45.87	0.600
66	2.0	15.28	47.56	3.41	23.86	0.23	23.71	35.71	0.611
75	2.0	15.28	46.51	3.04	21.73	0.23	24.78	34.12	0.613
67	2.0	28.85	64.57	5.05	25.15	0.4	39.41	44.86	0.601
76	2.0	28.88	63.29	4.52	21.92	0.41	41.38	42.61	0.604
77	2.0	38.41	74.53	5.46	22.52	0.55	52.01	48.52	0.597
68	2.0	38.53	74.56	5.90	24.85	0.54	49.71	49.71	0.596
78	2.0	61.79	99.26	7.13	23.22	0.86	76.04	61.24	0.583
69	2.0	62.29	98.45	8.55	25.52	0.82	72.93	61.98	0.582
84	3.0	19.11	48.12	2.73	22.80	0.28	25.32	35.46	0.611
81	3.0	28.85	58.33	3.48	22.96	0.42	35.39	40.65	0.606
80	3.0	61.86	88.94	6.00	23.13	0.86	65.81	56.04	0.589
79	4.0	61.85	82.10	5.31	23.24	0.86	58.86	52.67	0.593
83	4.5	59.34	77.73	4.78	23.27	0.81	54.37	50.55	0.595

Table A4. Mean Temperature of Rod and Cylinder
Surface at P = 1.0 Atm With Helium as
Test Gas.

Run No.	Rod Power (W)	Rod		Cylinder		T (°C)	T _f (°C)	C _r (Eq. (14))
		Temperature (°C)		Temperature (°C)				
		T _{mR}	Std. Dev.	T _{mC}	Std. Dev.			
48	5.02	28.77	0.93	22.6	0.1	6.17	25.67	0.622
50	10.01	35.22	1.79	22.87	0.15	12.35	29.05	0.618
49	14.84	41.02	2.52	23.02	0.19	18.0	32.02	0.615
47	19.85	47.31	3.61	23.35	0.23	23.96	35.33	0.611
51	28.16	58.38	4.55	25.06	0.34	33.32	41.72	0.605
46	39.1	69.84	6.08	25.18	0.44	44.66	47.51	0.598
44	43.74	77.58	6.66	28.5	0.48	49.08	53.04	0.592
42	48.86	81.52	7.37	27.08	0.54	54.44	54.3	0.591
41	54.24	87.24	8.1	27.66	0.6	59.58	57.45	0.587
40	63.33	95.63	9.28	27.19	0.7	68.45	61.41	0.583
39	73.86	107.51	10.60	29.57	0.81	77.94	68.54	0.575
38	84.26	116.99	11.82	30.0	0.91	86.99	73.5	0.570
37	98.96	129.99	13.47	30.42	1.06	99.57	80.21	0.562
36	107.83	135.99	14.68	27.78	1.17	107.71	81.64	0.561
35	119.15	144.27	15.9	27.07	1.3	117.2	85.67	0.556

Table A5. Mean Temperature of Rod and Cylinder
Surface With Helium as Test Gas.

Run No.	P (Atm)	Rod	Rod		Cylinder		T (°C)	T _f (°C)	C _r (Eq. (14))
		Power	Temperature (°C)		Temperature (°C)				
		(W)	T _{mR}	Std. Dev.	T _{mC}	Std. Dev.			
54	2.0	2.40	25.61	0.46	22.65	0.08	2.46	24.13	0.624
53	2.0	5.02	31.06	0.89	24.93	0.13	6.13	28.0	0.620
56	2.0	19.89	45.24	3.41	22.6	0.24	22.64	33.92	0.613
52	2.0	28.51	54.23	4.9	23.01	0.29	31.22	38.62	0.608
55	2.0	39.23	69.23	6.54	28.42	0.37	40.81	48.83	0.597
57	2.0	62.7	83.16	9.88	22.73	0.58	60.43	52.95	0.592
58	2.0	84.71	99.36	12.4	22.8	0.85	76.56	61.08	0.583
61	3.0	28.67	52.0	4.49	24.53	0.31	27.47	38.27	0.608
60	3.0	49.15	66.23	6.95	23.65	0.56	42.58	44.94	0.601
59	3.0	84.46	89.48	10.31	24.5	1.03	64.98	58.00	0.588
88	3.0	84.63	87.72	8.87	22.3	1.03	65.42	55.01	0.590
62	3.0	90.34	92.64	10.7	24.61	1.14	68.03	58.62	0.586
85	4.0	19.14	39.18	2.8	21.06	0.22	18.12	30.12	0.617
86	4.0	48.46	58.52	4.82	21.50	0.62	37.02	40.0	0.606
63	4.0	48.86	61.65	5.87	24.35	0.66	37.29	43.0	0.603
64	4.0	62.94	70.17	7.06	24.42	0.83	45.75	47.29	0.598
89	4.0	84.18	79.96	7.21	22.31	1.13	57.65	51.3	0.594
65	4.0	84.43	82.33	8.87	24.35	1.14	57.98	53.34	0.592
87	5.0	84.26	74.8	6.15	22.36	1.14	52.43	48.58	0.597

Table A6. Experimental Nusselt Numbers Versus
Rayleigh Numbers. Experimental Values
Are For Air as Test Gas and P = 1.0 Atm.

Rod Power (W)	Ra	Gr	Nu	Percent by Convection	Nu (total)
4.00	2.87x10 ⁴	4.08x10 ⁴	4.355	48.2	9.031
5.00	3.53x10 ⁴	5.02x10 ⁴	4.432	48.6	9.118
6.00	3.94x10 ⁴	5.58x10 ⁴	4.878	50.8	9.596
8.05	5.25x10 ⁴	7.45x10 ⁴	5.095	52.0	9.795
8.07	4.88x10 ⁴	6.92x10 ⁴	5.278	52.5	10.046
10.02	5.87x10 ⁴	8.32x10 ⁴	5.264	52.2	10.067
14.5	6.73x10 ⁴	9.55x10 ⁴	5.59	52.5	10.655
14.86	7.61x10 ⁴	1.08x10 ⁵	5.759	53.9	10.683
19.67	8.99x10 ⁴	1.27x10 ⁵	6.166	55.1	11.191
28.3	1.1 x10 ⁵	1.57x10 ⁵	6.485	55.5	11.692
29.32	1.01x10 ⁵	1.44x10 ⁵	6.434	54.5	11.805
39.44	1.29x10 ⁵	1.83x10 ⁵	6.559	54.6	12.01
43.82	1.32x10 ⁵	1.87x10 ⁵	7.305	57.1	12.795
48.50	1.32x10 ⁵	1.87x10 ⁵	7.445	57.0	13.064
53.99	1.4 x10 ⁵	1.98x10 ⁵	7.464	56.7	13.162
63.55	1.48x10 ⁵	2.1 x10 ⁵	7.113	54.6	13.024
73.62	1.53x10 ⁵	2.18x10 ⁵	7.206	54.2	13.291
83.65	1.56x10 ⁵	2.21x10 ⁵	7.423	54.3	13.666
98.61	1.59x10 ⁵	2.25x10 ⁵	7.526	53.7	14.017
108.25	1.59x10 ⁵	2.25x10 ⁵	7.626	53.5	14.264
119.34	1.63x10 ⁵	2.31x10 ⁵	7.98	54.2	14.717

Table A7. Experimental Nusselt Numbers Versus
Rayleigh Numbers. Experimental Values
Are For Air as Test.

Rod Power (W)	P (Atm)	Ra	Gr	Nu	Percent by Convection	Nu (total)
8.05	1.5	1.04×10^5	1.47×10^5	6.73	59.0	11.397
28.88	1.5	2.34×10^5	3.32×10^5	8.593	62.8	13.675
15.28	2.0	2.49×10^5	3.53×10^5	9.655	66.7	14.469
15.28	2.0	2.66×10^5	3.78×10^5	9.129	65.7	13.906
28.85	2.0	3.62×10^5	5.12×10^5	10.966	68.5	16.016
28.88	2.0	3.93×10^5	5.56×10^5	10.371	67.5	15.368
38.41	2.0	4.56×10^5	6.44×10^5	10.844	67.8	15.997
38.53	2.0	4.28×10^5	6.09×10^5	11.555	69.0	16.734
61.79	2.0	5.56×10^5	7.92×10^5	11.515	67.7	17.016
62.29	2.0	5.28×10^5	7.52×10^5	12.338	69.1	17.851
19.11	3.0	6.0×10^5	8.51×10^5	12.145	71.6	16.955
28.85	3.0	7.78×10^5	1.1×10^6	13.105	72.6	18.048
61.86	3.0	1.16×10^6	1.65×10^6	14.598	73.2	19.955
61.85	4.0	1.94×10^6	2.75×10^6	17.242	76.6	22.506
59.34	4.5	2.33×10^6	3.32×10^6	18.307	77.9	23.514

Table A8. Experimental Nusselt Numbers Versus
Rayleigh Numbers. Experimental Values
Are For Helium as Test Gas and P = 1.0 Atm.

Rod Power (W)	Ra	Gr	Nu	Percent by Convection	Nu (total)
5.02	2.47×10^2	4.22×10^2	2.843	75.7	3.281
10.01	5.66×10^2	8.03×10^2	2.44	75.0	3.255
14.84	7.91×10^2	1.12×10^3	2.462	74.8	3.293
19.85	1.0×10^3	1.42×10^3	2.441	74.2	3.29
28.16	1.28×10^3	1.81×10^3	2.431	73.4	3.312
39.1	1.58×10^3	2.25×10^3	2.481	73.1	3.395
43.74	1.62×10^3	2.29×10^3	2.478	72.4	3.421
48.86	1.76×10^3	2.5×10^3	2.486	72.3	3.436
54.24	1.85×10^3	2.63×10^3	2.497	72.1	3.465
63.33	2.02×10^3	2.87×10^3	2.505	71.7	3.496
73.86	2.1×10^3	2.98×10^3	2.504	70.8	3.534
84.26	2.2×10^3	3.13×10^3	2.521	70.4	3.581
98.96	2.32×10^3	3.29×10^3	2.531	69.7	3.63
107.83	2.47×10^3	3.5×10^3	2.537	69.6	3.647
119.15	2.56×10^3	3.63×10^3	2.542	69.1	3.677

Table A9. Experimental Nusselt Numbers Versus
Rayleigh Numbers. Experimental Values
Are For Helium as Test Gas.

Rod Power (W)	P (Atm)	Ra	Gr	Nu	Percent by Convection	Nu (total)
2.4	2.0	5.82×10^2	8.28×10^2	2.492	75.9	3.283
5.2	2.0	1.14×10^3	1.62×10^3	2.476	75.4	3.286
19.89	2.0	3.88×10^3	5.48×10^3	2.654	75.9	3.495
28.51	2.0	$5. \times 10^3$	7.08×10^3	2.736	76.0	3.602
39.23	2.0	5.68×10^3	8.08×10^3	2.799	75.3	3.718
62.7	2.0	7.96×10^3	1.13×10^4	3.038	76.3	3.983
84.71	2.0	9.08×10^3	1.29×10^4	3.193	76.3	4.184
28.67	3.0	9.99×10^3	1.41×10^4	3.261	79.1	4.125
49.15	3.0	1.4×10^4	1.99×10^4	3.612	80.0	4.513
84.46	3.0	1.83×10^4	2.59×10^4	3.988	80.5	4.955
84.63	3.0	1.89×10^4	2.68×10^4	3.986	80.6	4.943
90.34	3.0	1.87×10^4	2.66×10^4	4.064	80.6	5.04
19.14	4.0	1.31×10^4	1.86×10^4	3.412	80.6	4.233
48.46	4.0	2.34×10^4	3.3×10^4	4.272	83.0	5.146
48.86	4.0	2.26×10^4	3.2×10^4	4.232	82.6	5.121
62.94	4.0	2.61×10^4	3.7×10^4	4.422	82.9	5.334
84.18	4.0	3.12×10^4	4.42×10^4	4.685	83.4	5.62
84.43	4.0	3.04×10^4	4.32×10^4	4.636	83.-	5.582
84.26	5.0	4.58×10^4	6.5×10^4	5.295	85.2	6.215



APPENDIX B

ANALYSIS OF EXPERIMENTAL ERROR



Analysis of Experimental Errors

The Rayleigh number and Nusselt number are subject to both accidental and systematic errors. The latter results because of uncertainties in the parameters and quantities used to calculate both dimensionless numbers.

In the present study, accidental error was essentially eliminated by taking two repeated measurements and then arithmetically averaging the result.

Systematic error includes uncertainties in the thermophysical properties of the test gas, geometrical dimensions, temperature and power measurements and power lost through the end plates of the cylinder. Generally, the law of summation of fractional errors [24] can be applied to give an estimate of the total fractional uncertainty in the present case where uncertainties of component quantities are easily estimated. For the Rayleigh number, one has

$$f_{Ra} = f_{\text{properties}} + f_{\text{geometric}} + f_{\text{temperature}}$$

and for the Nusselt number,

$$f_{Nu} = f_{\text{properties}} + f_{\text{geometric}} + f_{\text{power}} + f_{\text{temperature}}$$

Table B1 contains a summary of the uncertainties of above mentioned quantities.

The combined uncertainties in the thermophysical properties, geometrical factors and temperature measurements produce a total uncertainty of 6% to 8% in the Rayleigh number. Similarly, the total uncertainty in the Nusselt number is 8% to 10%.

Table B1. Experimental Uncertainties of Quantities
Used to Compute Ra and Nu.

<u>Quantity or Parameter</u>	<u>Fractional Uncertainty+</u>
Thermophysical Properties	0.02
Length Dimensions	0.01
Temperature	0.015++
Power	0.005 (measurement)
	0.03 (end loss)

+ Maximum value obtained either by direct measurement or a combination of literature values, e.g., thermophysical properties.

++Maximum value obtained by combining standard errors of the thermocouple wire calibration and millivolt readings.

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